ASHRAE Official Duct Fitting Database

Master Engineering Catalog & Full Process Flow Architecture

Complete Technical Reference for All 234 Verified Duct Fittings across Supply, Common, and Exhaust Systems

234	3	3	12
OFFICIAL FITTINGS	FLOW SYSTEMS	DUCT GEOMETRIES	CATEGORIES
100% Equation-Verified	Supply, Common, Exhaust	Round, Rectangular, Oval	Elbows, Tees, Wyes, Plenums

Document Overview & Purpose: This comprehensive reference document represents the fully expanded hierarchical catalog of the official ASHRAE Duct Fitting Database (ASHRAE Fundamentals, Chapter 21: Duct Design). It provides the complete process flow and technical data for all **234 official duct fittings** exactly as revealed upon triggering **'Expand All'** within the DuctFit Pro engineering analysis suite. Each fitting is fully characterized with its formal classification, complete hierarchy breadcrumb, authoritative peer-reviewed research citation, high-resolution CAD blueprint diagram, comprehensive input parameter definitions (in dual SI and IP unit systems with validated physical limits), output aerodynamics (including velocity, velocity pressure, dimensionless local loss coefficient, and highlighted pressure loss), and official governing equations evaluated by the engine.

Distribution Flow System	Round Duct	Rectangular Duct	Flat Oval Duct	System Total	Primary Functional Scope
Supply Air Systems	43 fittings (SD)	39 fittings (SR)	6 fittings (SF)	**88 fittings**	Diverging junctions (wyes, tees, crosses), air distribution exits, diffuser cones, fan outlets
Common Systems (Supply & Return)	39 fittings (CD)	41 fittings (CR)	7 fittings (CF)	**87 fittings**	Straight duct runs (Colebrook), elbows (smooth & mitered with vanes), obstructions, dampers, silencers
Exhaust / Return Systems	33 fittings (ED)	23 fittings (ER)	3 fittings (EF)	**59 fittings**	Converging junctions (intake tees & wyes), suction entries, bellmouths, kitchen/canopy hoods
Total Verified Catalog	**111 Round**	**108 Rectangular**	**15 Flat Oval**	**234 FITTINGS**	**100% Equation & Table Parity with Official ASHRAE Database**

Authoritative Standard: ASHRAE Fundamentals, Chapter 21 (Duct Design)	**Software Build:** DuctFit Pro Version v1.0.8
Database Engine: Official 2014 ASHRAE Duct Fitting Database (Jet OLEDB 2014 Synchronization)	**Publication Date:** 29-October-2026
Verification Status: 234 of 234 fittings verified with 0 calculation exceptions (SI & IP)	**Unit Parity: Dual System (SI Metric / IP Inch-Pound)**

System Process Flow & Hierarchy Architecture

The following diagrams illustrate the exact hierarchical distribution tree connecting systems, duct geometries, and aerodynamic functional subcategories down to individual fitting groups as structured in the database.

SUPPLY AIR SYSTEMS **Total: 88 Fittings**	**ROUND DUCT (SD)** **43 Fittings**	• **Plenums:** SD1-1, SD1-2 • **Exits:** SD2-1 to SD2-7 • **Transitions:** SD4-1, SD4-2 • **Diverging Junctions:** - Wye (Conical Branch): SD5-1 to SD5-4 - Wye (Non-Conical): SD5-5 to SD5-10 - Tee (Conical Branch): SD5-11 to SD5-14 - Tee (Non-Conical): SD5-15 to SD5-19 - Tee (Bullhead): SD5-20, SD5-21 - Cross: SD5-22 to SD5-25 • **Fan & System Interactions:** SD7-1 to SD7-5
	RECTANGULAR (SR) **39 Fittings**	• **Plenums:** SR1-1 • **Exits:** SR2-1 to SR2-6 • **Elbows:** SR3-1 • **Transitions:** SR4-1 to SR4-3 • **Diverging Junctions:** - Wye (Symmetrical): SR5-1 - Wye (Unsymmetrical): SR5-2, SR5-3 - Tee: SR5-4 to SR5-16 • **Fan Interactions:** SR7-1 to SR7-15
	FLAT OVAL (SF) **6 Fittings**	• **Transitions:** SF4-1 • **Diverging Junctions:** - Wye (Conical): SF5-1 - Wye (Non-Conical): SF5-2 - Tee (Conical): SF5-3 - Tee (Non-Conical): SF5-4

COMMON SYSTEMS **Total: 87 Fittings**	**ROUND DUCT (CD)** **39 Fittings**	• **Straight Duct:** CD11-1 (Standard Darcy-Colebrook), CD11-2 (Flexible Duct) • **Elbows (Bends):** - Mitered (No Vanes / Vanes): CD3-1 to CD3-6 - Die Stamped: CD3-7 to CD3-10 - Pleated: CD3-11, CD3-12 - Gored: CD3-13 to CD3-20 • **Obstructions:** CD6-1 to CD6-3 (Pipes & Beams) • **Duct Equipment:** CD8-1 to CD8-8 (Double/Single Wall Silencers) • **Dampers:** CD9-1 to CD9-4 (Butterfly & Gate Dampers)
	RECTANGULAR (CR) **41 Fittings**	• **Straight Duct:** CR11-1 (Darcy-Colebrook Equivalent Rectangular) • **Elbows (Bends):** - Smooth Radius: CR3-1 to CR3-5 - Mitered with Vanes: CR3-6 to CR3-14 (Single/Double Thickness Vanes) - Other Specialized: CR3-15 to CR3-18 • **Obstructions:** CR6-1 to CR6-4 (Internal Rods & Beams) • **Duct Equipment:** CR8-1 to CR8-11 (Silencers, Heating/Cooling Coils, VAV Boxes) • **Dampers:** CR9-1 to CR9-14 (Opposed & Parallel Blade Volume/Fire Dampers)
	FLAT OVAL (CF) **7 Fittings**	• **Straight Duct:** CF11-1 (Heyt-Diaz Equivalent Flat Oval) • **Elbows:** CF3-1 to CF3-6 (Major & Minor Axis Smooth Radius Bends)

EXHAUST / RETURN **Total: 59 Fittings**	**ROUND DUCT (ED)** **33 Fittings**	• **Entries / Inlets:** ED1-1 to ED1-8 (Bellmouth & Flush Suction Inlets) • **Plenums:** ED1-4 • **Transitions:** ED4-1, ED4-2 • **Converging Junctions:** - Wye: ED5-1 to ED5-4 - Tee: ED5-5 to ED5-8 - Bullhead: ED5-9 • **Fan Interactions:** ED7-1 to ED7-8 (Centrifugal, Tubeaxial, Vaneaxial Inlets) • **Hoods:** ED10-1 (Hood, Conical, Round Duct), ED10-2 (Hood, Rectangular-Shaped, Tapered, Round Duct)
	RECTANGULAR (ER) **23 Fittings**	• **Entries:** ER1-1 to ER1-3 • **Plenums:** ER1-4 to ER1-7 • **Elbows:** ER3-1 • **Transitions:** ER4-1 to ER4-3 • **Converging Junctions:** - Wye: ER5-1 to ER5-5 - Tee: ER5-6 to ER5-10 • **Fan Interactions:** ER7-1 to ER7-3 (SWSI Centrifugal Inlets)
	FLAT OVAL (EF) **3 Fittings**	• **Transitions:** EF4-1 • **Converging Junctions:** - Tee (Straight Body Branch): EF5-1 - Wye (45° Straight Body Branch): EF5-3

Master Fitting Cross-Reference Directory

Complete index of all 234 official ASHRAE duct fittings sorted by catalog code, summarizing classification, flow system, geometry, functional category, and authoritative research citation.

Code	Official Fitting Title / Description	System	Shape	Category	Research Citation
CD11-1	Straight Duct, Round	Common	Round	Round	Colebrook 1939
CD11-2	Straight Duct, Flexible	Common	Round	Round	Abushakra 2004 and Culp 2011
CD3-1	Elbow, Die Stamped, 90 Degree, r/D = 1.5	Common	Round	Die Stamped	UMC 1985, Report SRF785
CD3-10	Elbow, 7 Gore, 90 Degree, r/D = 2.5	Common	Round	Gored	UMC 1985, Report SRF1285
CD3-11	Elbow, Flat-back, 90 Degree (UMC 1985, Report SRF1285)	Common	Round	Elbows	
CD3-12	Elbow, 3 Gore, 90 Degree, r/D=0.75 to 2.0	Common	Round	Gored	Locklin 1950, Fig. 10
CD3-13	Elbow, 3 Gore, 60 Degree, r/D = 1.5	Common	Round	Gored	[UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)]
CD3-14	Elbow, 3 Gore, 45 Degree, r/D = 1.5	Common	Round	Gored	[UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)]
CD3-15	Elbow, Mitered, 90 Degree	Common	Round	Mitered	UMC 1985, Report SRF1285
CD3-16	Elbow, Mitered, 60 Degree	Common	Round	Mitered	[UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1), Graph (a)]
CD3-17	Elbow, Mitered, 45 Degree	Common	Round	Mitered	[UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)]
CD3-18	Elbow, Mitered, 30 Degree	Common	Round	Mitered	[UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1 (Graph (a)]
CD3-19	Elbow, Mitered, 22.5 Degree	Common	Round	Mitered	[UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)]
CD3-2	Elbow, Die Stamped, 90 Degree, r/D = 1.0	Common	Round	Die Stamped	UMC 1985, Report SRF785
CD3-20	Elbow, Mitered with Vanes, 90 Degree	Common	Round	Mitered	UMC 1985, Report SRF1285
CD3-21	Elbow, 5 Gore, 90 Degree, r/D = 1.0	Common	Round	Gored	UMC 1985, Report SRF1285; Brooks 1993
CD3-3	Elbow, Die Stamped, 45 Degree, r/D = 1.5	Common	Round	Die Stamped	[UMC 1985, Report SRF785; Idelchik 1986, Diagram 6-1, Graph (a)]
CD3-4	Elbow, Die Stamped, 45 Degree, r/D = 1.0	Common	Round	Die Stamped	[UMC 1985, Report SRF785; Idelchik 1986, Diagram 6-1, Graph (a)]
CD3-5	Elbow, Pleated, 90 Degree, r/D = 1.5	Common	Round	Pleated	UMC 1989, Report SRS189
CD3-7	Elbow, Pleated, 45 Degree, r/D = 1.5	Common	Round	Pleated	UMC 1989, Report SRS189
CD3-9	Elbow, 5 Gore, 90 Degree, r/D = 1.5	Common	Round	Gored	UMC 1985, Report SRF1285
CD6-1	Screen (Only)	Common	Round	Obstructions	Idelchik 1986, Diagrams 8-6 and 3-15
CD6-2	Perforated Plate, Thick	Common	Round	Obstructions	Idelchik 1986, Diagram 8-3
CD6-3	Obstruction, Smooth Cylinder in Round Duct	Common	Round	Obstructions	Idelchik 1986, Diagram 10-1
CD8-1	Silencer, Ducted, Double Wall with Bullet	Common	Round	Silencer, Double Wall	Standard (High) Pressure Drop (see Table below
CD8-2	Silencer, Ducted, Double Wall with Bullet and Tail Cone	Common	Round	Silencer, Double Wall	Standard (High) Pressure Drop (see Table below
CD8-3	Silencer, Ducted, Double Wall with Bullet	Common	Round	Silencer, Double Wall	Low Pressure Drop (see Table below
CD8-4	Silencer, Ducted, Double Wall with Bullet and Tail Cone	Common	Round	Silencer, Double Wall	Low Pressure Drop (see Table below
CD8-5	Silencer, Ducted, Single Wall with Bullet	Common	Round	Silencer, Single Wall	Standard (High) Pressure Drop
CD8-6	Silencer, Ducted, Single Wall with Bullet and Tail Cone	Common	Round	Silencer, Single Wall	Standard (High) Pressure Drop
CD8-7	Silencer, Ducted, Single Wall with Bullet	Common	Round	Silencer, Single Wall	Low Pressure Drop (see Table below
CD8-8	Silencer, Ducted, Single Wall with Bullet and Tail Cone	Common	Round	Silencer, Single Wall	Low Pressure Drop (see Table below
CD9-1	Damper, Butterfly	Common	Round	Dampers	Idelchik 1986, Diagram 9-16
CD9-2	Damper, Gate	Common	Round	Dampers	Idelchik 1986, Diagram 9-5
CD9-3	Fire Damper, Curtain Type, Style C, Horizontal Duct (Vertical Installation)	Common	Round	Dampers	1-1/2 && 3 hour rated; For use in dynamic && static systems
CD9-4	Fire Damper, Curtain Type, Style C, Vertical Duct (Horizontal Installation)	Common	Round	Dampers	1-1/2 && 3 hour rated; For use in dynamic && static systems

Code	Official Fitting Title / Description	System	Shape	Category	Research Citation
CF11-1	Straight Duct, Flat Oval	Common	Flat Oval	Flat Oval	Heyt and Diaz 1975; Townsend et al. 1994
CF3-1	Elbow, Flat Oval, 5 Gore, Easy Bend, 90 Degree	Common	Flat Oval	Elbows	Kulkarni, Khaire and Idem 2009
CF3-2	Elbow, Flat Oval, 5 Gore, Hard Bend, 90 Degree	Common	Flat Oval	Elbows	Townsend et al. 1996 and Smith && Jones 1976
CF3-3	Elbow, Flat Oval, Mitered, Easy Bend (No Vanes), 90 Degree	Common	Flat Oval	Elbows	Kulkarni, Khaire and Idem 2009
CF3-4	Elbow, Flat Oval, Mitered, Hard Bend (No Vanes), 90 Degree	Common	Flat Oval	Elbows	Kulkarni, Khaire and Idem 2009
CF3-5	Elbow, Flat Oval, Mitered, Easy Bend with Vanes, 90 Degree	Common	Flat Oval	Elbows	Kulkarni, Khaire and Idem 2009
CF3-6	Elbow, Flat Oval, Mitered, Hard Bend with Vanes, 90 Degree	Common	Flat Oval	Elbows	Kulkarni, Khaire and Idem 2009
CR11-1	Straight Duct, Rectangular	Common	Rectangular	Rectangular	Huebscher 1948
CR3-1	Elbow, Smooth Radius without Vanes	Common	Rectangular	Smooth Radius	Idelchik 1986, Diagram 6-1
CR3-12	Elbow, Mitered, 90 Degree, Single-Thickness Vanes	Common	Rectangular	Single-Thickness Vanes	3 1/4-in. Vane Spacing (Brooks 1993
CR3-14	Elbow, Mitered, 90 Degree, Double-Thickness Vanes	Common	Rectangular	Double-Thickness Vanes	1 1/2-in. Vane Spacing (Brooks 1993
CR3-15	Elbow, Mitered, 90 Degree, Double-Thickness Vanes	Common	Rectangular	Double-Thickness Vanes	2 1/8-in. Vane Spacing (Rozell 1974
CR3-16	Elbow, Mitered, 90 Degree, Double-Thickness Vanes	Common	Rectangular	Double-Thickness Vanes	3 1/4-in. Vane Spacing (Brooks 1993
CR3-17	Elbow, Z-Shape	Common	Rectangular	Other Elbows	Idelchik 1986, Diagram 6-11
CR3-2	Elbow, Sharp Throat (r/W=0.5), Radius Heel, 90 Degree	Common	Rectangular	Other Elbows	Idelchik 1986, Diagram 6-1
CR3-3	Elbow, Smooth Radius, One Splitter Vane	Common	Rectangular	Smooth Radius	Locklin 1950; Madison && Parker 1936
CR3-4	Elbow, Smooth Radius, Two Splitter Vanes	Common	Rectangular	Smooth Radius	Locklin 1950; Madison && Parker 1936
CR3-5	Elbow, Smooth Radius, Three Splitter Vanes	Common	Rectangular	Smooth Radius	Locklin 1950; Madison && Parker 1936
CR3-6	Elbow, Mitered (Idelchik 1986, Diagram 6-5)	Common	Rectangular	Mitered	
CR3-7	Elbow, Rounded Throat, Sharp Heel, 90 Degree	Common	Rectangular	Other Elbows	Idelchik 1986, Diagram 6-8
CR3-8	Elbow, Beveled Throat, Sharp Heel, 90 Degree	Common	Rectangular	Other Elbows	Idelchik 1986, Diagram 6-8
CR3-9	Elbow, Mitered, 90 Degree, Single-Thickness Vanes	Common	Rectangular	Single-Thickness Vanes	1 1/2-in. Vane Spacing (Brooks 1993
CR6-1	Screen (Only) (Idelchik 1986, Diagrams 8-6 and 3-15)	Common	Rectangular	Obstructions	
CR6-2	Perforated Plate, Thick	Common	Rectangular	Obstructions	Idelchik 1986, Diagram 8-3
CR6-3	Orifice, Thick-Edged	Common	Rectangular	Obstructions	Idelchik 1986, Diagram 4-15
CR6-4	Obstruction, Smooth Cylinder in Rectangular Duct	Common	Rectangular	Obstructions	Idelchilk 1986, Diagram 10-1
CR8-1	Silencer, Ducted, Standard Pressure Drop (see Table below)	Common	Rectangular	Silencer	
CR8-10	Terminal Control Unit, Basic Assembly, One-Row Coil and Attenuator	Common	Rectangular	Terminal Unit (Box)	
CR8-11	Terminal Control Unit, Basic Assembly, Two-Row Coil and Attenuator	Common	Rectangular	Terminal Unit (Box)	
CR8-2	Silencer, Ducted, Medium Pressure Drop (see Table below)	Common	Rectangular	Silencer	
CR8-3	Silencer, Ducted, Medium-Low Pressure Drop (see Table below)	Common	Rectangular	Silencer	
CR8-4	Silencer, Ducted, Low Pressure Drop (see Table below)	Common	Rectangular	Silencer	
CR8-5	Coil, Electric, Open Coil	Common	Rectangular	Coil	
CR8-6	Coil, Electric, Finned Tubular	Common	Rectangular	Coil	
CR8-7	Coil, Hot Water, One Row	Common	Rectangular	Coil	
CR8-8	Coil, Hot Water, Two Row	Common	Rectangular	Coil	
CR8-9	Terminal Control Unit, Basic Assembly and Attenuator	Common	Rectangular	Terminal Unit (Box)	
CR9-1	Damper, Butterfly (Idelchik 1986, Diagram 9-17)	Common	Rectangular	Dampers	
CR9-10	Fire Damper, Curtain Type, Style C, Vertical Duct (Horiontal Installation)	Common	Rectangular	Dampers	1-1/2 && 3 Hour; For Use in Dynamoc && Static Systems
CR9-11	Fire/Smoke Damper, 3 Hour, Airfoil Blades, Vertical Installation	Common	Rectangular	Dampers	Van Becelaere && Sauer 2004
CR9-12	Fire/Smoke Damper, 1-1/2 Hour, 3V Blades, Vertical Installation	Common	Rectangular	Dampers	Van Becelaere && Sauer 2004
CR9-13	Fire/Smoke Damper, 3 Hour, Airfoil Blades, Horizontal Installation	Common	Rectangular	Dampers	Van Becelaere && Sauer 2004

Code	Official Fitting Title / Description	System	Shape	Category	Research Citation
CR9-14	Fire/Smoke Damper, 1-1/2 Hour, 3V Blades, Horizontal Installation	Common	Rectangular	Dampers	Van Becelaere && Sauer 2004
CR9-2	Damper, Gate	Common	Rectangular	Dampers	Idelchik 1986, Diagram 9-5
CR9-3	Damper, Parallel && Opposed 3V Blades, Open	Common	Rectangular	Dampers	Van Becelaere && Sauer 2004
CR9-4	Damper, Parallel && Opposed Airfoil Blades, Open	Common	Rectangular	Dampers	Van Becelaere && Sauer 2004
CR9-5	Fire Damper, Curtain Type, Style A, Horizontal Duct (Vertical Installation)	Common	Rectangular	Dampers	1-1/2 && 3 Hour; For Use in Dynamic && Static Systems
CR9-6	Fire Damper, Curtain Type, Style B, Horizontal Duct (Vertical Installation)	Common	Rectangular	Dampers	1-1/2 && 3 Hour; For Use in Dynamic && Static Systems
CR9-7	Fire Damper, Curtain Type, Style C, Horizontal Duct (Vertical Installation)	Common	Rectangular	Dampers	1-1/2 && 3 Hour; For Use in Dynamic && Static Systems
CR9-8	Fire Damper, Curtain Type, Style A, Vertical Duct (Horizontal Installation)	Common	Rectangular	Dampers	1-1/2 && 3 Hour; For Use in Dynamic && Static Systems
CR9-9	Fire Damper, Curtain Type, Style B, Vertical Duct (Horizontal Installation)	Common	Rectangular	Dampers	1-1/2 && 3 Hour; For Use in Dynamic && Static Systems
ED1-1	Duct Mounted in Wall	Exhaust/Return	Round	Entries	Idelchik 1986, Diagram 3-1
ED1-2	Bellmouth, Standard Radius	Exhaust/Return	Round	Entries	Idelchik 1986, Diagram 3-4
ED1-3	Bellmouth (Standard Radius) with Wall	Exhaust/Return	Round	Entries	Idelchik 1986, Diagram 3-4
ED1-4	Conical Bellmouth/Sudden Contraction	Exhaust/Return	Round	Entries	Idelchik 1986, Diagram 3-6
ED1-5	Conical Bellmouth/Sudden Contraction, with Wall	Exhaust/Return	Round	Entries	Idelchik 1986, Diagram 3-7
ED1-6	Intake Hood	Exhaust/Return	Round	Entries	Idelchik 1986, Diagram 3-18
ED1-7	Orifice, Sharp-Edged	Exhaust/Return	Round	Entries	Brooks 1993
ED1-8	Bellmouth, Short Radius	Exhaust/Return	Round	Entries	Idelchik 1986, Diagram 3-4
ED10-1	Hood, Conical, Round Duct	Exhaust/Return	Round	Hoods	Brandt 1946
ED10-2	Hood, Rectangular-Shaped, Tapered, Round Duct	Exhaust/Return	Round	Hoods	Brandt 1946
ED2-1	Conical Diffuser, Round to Plenum, Exhaust/Return Systems	Exhaust/Return	Round	Plenums	Idelchik 1986, Diagram 5-8
ED4-1	Transition, Round to Round, Exhaust/Return Systems	Exhaust/Return	Round	Transitions	Idelchik 1986, Diagrams 5-2 && 5-22
ED4-2	Transition, Round to Rectangular, Exhaust/Return Systems	Exhaust/Return	Round	Transitions	Idelchik 1986, Diagram 5-27
ED5-1	Wye, 30 Degree, Converging	Exhaust/Return	Round	Wye	Sepsy 1973
ED5-10	Double Wye, 45 Degree, Ab1 = Ab2, As = Ac, Converging	Exhaust/Return	Round	Junctions, Converging	Idelchik 1986, Diagram 7-27
ED5-11	Cross, Ab1 = Ab2, As = Ac, Converging	Exhaust/Return	Round	Junctions, Converging	Idelchik 1986, Diagram 7-29
ED5-2	Wye, 45 Degree, Converging	Exhaust/Return	Round	Wye	Sepsy 1973
ED5-3	Tee, Converging	Exhaust/Return	Round	Tee	Sepsy 1973
ED5-4	Bullhead Tee without Vanes, Db1 > or = Db2, Converging	Exhaust/Return	Round	Bullhead	UMC 1985, Report SRF785E
ED5-6	Capped Wye, Branch with 45 Degree Elbow, Branch 90 Degrees	Exhaust/Return	Round	Wye	to Main, Converging (UMC 1983, Report SRN1083
ED5-7	Capped Tee, Converging	Exhaust/Return	Round	Tee	UMC 1983, Report SRN1083
ED5-8	Heel-Tapped Elbow, 90 Degree, r/Dc = 1.5, Converging	Exhaust/Return	Round	Junctions, Converging	UMC 1985, Report SRF785E
ED5-9	Symmetrical Wye, 60 Degree, Db1 > or = Db2, Converging	Exhaust/Return	Round	Wye	UMC 1985, Report SRF785E
ED7-1	Centrifugal Fan Located in Plenum or Cabinet	Exhaust/Return	Round	Fan & System Interactions	AMCA 1990, Fig. 9-11
ED7-2	Fan Inlet, Centrifugal, SWSI, with 4-Gore Elbow	Exhaust/Return	Round	Fan Inlet, Centrifugal	AMCA 1990, Fig. 9-5
ED7-3	Fan Inlet, Centrifugal, SWSI, with 3-Gore Elbow	Exhaust/Return	Round	Fan Inlet, Centrifugal	AMCA 1990, Fig. 9-5
ED7-4	Fan Inlet, Centrifugal, SWSI, with Mitered Elbow	Exhaust/Return	Round	Fan Inlet, Centrifugal	AMCA 1990, Fig. 9-5
ED7-5	Fan Inlet, Tubeaxial, 0.25 Hub-Tip Ratio, with 4-Gore Elbow	Exhaust/Return	Round	Fan Inlet, Tubeaxial	AMCA 1990, Fig. 9-2
ED7-6	Fan Inlet, Tubeaxial, 0.25 Hub-Tip Ratio, with Mitered Elbow	Exhaust/Return	Round	Fan Inlet, Tubeaxial	AMCA 1990, Fig. 9-2
ED7-7	Fan Inlet, Tubeaxial, 0.35 Hub-Tip Ratio, with Mitered Elbow	Exhaust/Return	Round	Fan Inlet, Tubeaxial	AMCA 1990, Fig. 9-2
ED7-8	Fan Inlet, Vaneaxial, 0.61 Hub-Tip Ratio, with 4-Gore Elbow	Exhaust/Return	Round	Fan Inlet, Vaneaxial	AMCA 1990, Fig. 9-2
ED7-9	Fan Inlet, Vaneaxial, 0.61 Hub-Tip Ratio, with Mitered Elbow	Exhaust/Return	Round	Fan Inlet, Vaneaxial	AMCA 1990, Fig. 9-2
EF4-1	Transition, Flat Oval, Concentric && Eccentrix, Exhaust/Return Air Systems	Exhaust/Return	Flat Oval	Transitions	Zhang, Sun, Ford && Denny 2014
EF5-1	Tee, Flat Oval, Straight Body Branch, Converging	Exhaust/Return	Flat Oval	Junctions, Converging	Kulkarni, Cui && Idem 2011
EF5-3	Wye, Flat Oval, 45 Degree, Straight Body Branch, Converging	Exhaust/Return	Flat Oval	Junctions, Converging	Kulkarni, Cui && Idem 2011
ER1-1	Duct Mounted in Wall (Idelchik 1986, Diagram 3-1)	Exhaust/Return	Rectangular	Entries	

Code	Official Fitting Title / Description	System	Shape	Category	Research Citation
ER1-2	Conical Bellmouth/Sudden Contraction	Exhaust/Return	Rectangular	Entries	Idelchik 1986, Diagram 3-6
ER1-3	Conical Bellmouth/Sudden Contraction, with Wall	Exhaust/Return	Rectangular	Entries	Idelchik 1986, Diagram 3-7
ER2-1	Bellmouth, Plenum to Round, Exhaust/Return Systems	Exhaust/Return	Rectangular	Plenums	Idelchik 1986, Diagram 4-9
ER2-2	Conical Bellmouth/Sudden Contraction, Plenum to Round	Exhaust/Return	Rectangular	Plenums	Exhaust/Return Systems (Idelchik 1986, Diagram 4-9
ER2-3	Plain Diffuser (Two Sides Parallel), Rectangular to Plenum	Exhaust/Return	Rectangular	Plenums	Exhaust/Return Systems (Idelchik 1986, Diagram 5-10
ER2-4	Pyramidal Diffuser, Rectangular to Plenum	Exhaust/Return	Rectangular	Plenums	Exhaust/Return Systems (Idelchik 1986, Diagram 5-9
ER3-1	Elbow, 90 Degree, Variable Inlet/Outlet Areas	Exhaust/Return	Rectangular	Elbows	Exhaust/ Return Systems (Idelchik 1986, Diagram 6-4
ER4-1	Transition, Rectangular, Two Sides Parallel, Symmetrical	Exhaust/Return	Rectangular	Transitions	Exhaust/Return Systems (Idelchik 1986, Diagram 5-5
ER4-2	Transition, Pyramidal, Exhaust/Return Systems	Exhaust/Return	Rectangular	Transitions	Idelchik 1986, Diagram 5-4
ER4-3	Transition, Rectangular to Round, Exhaust/Return Systems	Exhaust/Return	Rectangular	Transitions	Idelchik 1986, Diagram 5-27
ER5-1	Smooth Radius Wye of the Type As + Ab > or = Ac, Branch 90	Exhaust/Return	Rectangular	Wye	Degrees to Main, Converging (Idelchik 1986, Diagram 7-11
ER5-10	Cross of the Type As = Ac, Ab1 = Ab2, Converging	Exhaust/Return	Rectangular	Junctions, Converging	Idelchik 1986, Diagram 7-29
ER5-2	Tee, Round Tap to Rectangular Main [Ab/As=Ab/Ac=0.5; As/Ac], Converging	Exhaust/Return	Rectangular	Tee	SMACNA 1990, Sepsy 1973
ER5-3	Tee, 45 Degree Entry Branch [Ab/As=Ab/Ac=0.5; As=Ac], Converging	Exhaust/Return	Rectangular	Tee	SMACNA 1990, Sepsy 1973
ER5-4	Wye, Symmetrical, Dovetail, Qb/Qc = 0.5, Converging	Exhaust/Return	Rectangular	Wye	Idelchik 1986, Diagram 7-24
ER5-5	Bullhead Tee without Vanes, Converging	Exhaust/Return	Rectangular	Tee	Idelchik 1986, Diagram 7-23
ER5-6	Symmetrical Wye Ab1 + Ab2 = Ac, Ab1 = Ab2, 30 Degree, Converging	Exhaust/Return	Rectangular	Wye	Idelchik 1986, Diagram 7-30
ER5-7	Symmetrical Wye Ab1 + Ab2 = Ac, Ab1 = Ab2, 60 Degree, Converging	Exhaust/Return	Rectangular	Wye	Idelchik 1986, Diagram 7-30
ER5-8	Symmetrical Wye of the Type Ab1 + Ab2 = Ac, Ab1 = Ab2	Exhaust/Return	Rectangular	Wye	90 Degree, Converging (Idelchik 1986, Diagrams 7-30
ER5-9	Double Wye of the Type As = Ac, Ab1 = Ab2, 45 Degree, Converging	Exhaust/Return	Rectangular	Junctions, Converging	Idelchik 1986, Diagram 7-27
ER7-1	Fan Inlet, Centrifugal, SWSI, 90 Degree Smooth Radius Elbow	Exhaust/Return	Rectangular	Fan Inlet, Centrifugal, SWSI	Square) (AMCA 1990, Fig. 9-6
ER7-2	Fan Inlet, Centrifugal, SWSI, 90 Deg. Smooth Radius Elbow	Exhaust/Return	Rectangular	Fan Inlet, Centrifugal, SWSI	Square) with Fulll Radius Vanes (ACMA 1990, Fig. 9-6
ER7-3	Fan Inlet, Centrifugal, SWSI, 90 Degree Smooth Radius Elbow	Exhaust/Return	Rectangular	Fan Inlet, Centrifugal, SWSI	Square) with Short Vanes (AMCA 1990, Fig. 9-6
SD1-1	Bellmouth (Standard Radius), Plenum to Round, Supply Air Systems	Supply	Round	Plenums	Idelchik 1986, Diagram 4-9 and 3-4
SD1-2	Conical Bellmouth/Sudden Contraction, Plenum to Round	Supply	Round	Plenums	Supply Air Systems (Idelchik 1986, Diagram 4-9
SD2-1	Abrupt Exit	Supply	Round	Exits	Idelchik 1986, Diagram 11-1
SD2-2	Discharge from a Duct Flush with Wall	Supply	Round	Exits	Idelchik 1986, Diagram 11-2
SD2-3	Conical Diffuser, Free Discharge	Supply	Round	Exits	Idelchik 1986, Diagram 11-3
SD2-4	Conical Diffuser, with Wall	Supply	Round	Exits	Idelchik 1986, Diagram 5-8
SD2-5	Exhaust Hood	Supply	Round	Exits	Idelchik 1986, Diagram 11-16
SD2-6	Stackhead (Dexit/D=1)	Supply	Round	Exits	Idelchik 1986, Diagram 11-23
SD2-7	Stackhead (Dexit/D<1)	Supply	Round	Exits	Idelchik 1986, Diagram 11-23
SD4-1	Transition, Round to Round, Supply Air Systems	Supply	Round	Transitions	Idelchik 1986, Diagram 5-2 && 5-22
SD4-2	Transition, Rectangular to Round, Supply Air Systems	Supply	Round	Transitions	Idelchik 1986, Diagram 5-27
SD5-1	Wye, 45 Degree, Diverging	Supply	Round	Non-Conical Branch	UMC 1986, Report SRF386
SD5-10	Tee, Conical Branch Tapered into Body, Diverging	Supply	Round	Conical Branch	UMC 1986, Report SRF386
SD5-11	Tee, Conical Branch, Diverging	Supply	Round	Conical Branch	Jones 1969, Fig. 12; UMC 1986, Report SRF386
SD5-12	Tee, 45 Degree Entry Branch, Diverging	Supply	Round	Non-Conical Branch	UMC 1986, Report SRF386
SD5-13	Tee, Branch with 45 Degree Elbow, Branch 90 Degrees to Main, Diverging	Supply	Round	Non-Conical Branch	Jones 1969, Fig. 18; UMC 1986, Report SRF386
SD5-14	Tee, Conical Branch with 45 Deg. Elbow, Branch 90 Deg. to Main, Diverging	Supply	Round	Conical Branch	Jones 1969, Fig. 19; UMC 1986, Report SRF386
SD5-15	Tee, Branch with 90 Degree Elbow, Branch 90 Degrees to Main, Diverging	Supply	Round	Non-Conical Branch	Jones 1969, Fig. 17; UMC 1986, Report SRF386
SD5-16	Tee, Branch with 60 Degree Elbow, Branch 45 Degrees to Main, Diverging	Supply	Round	Non-Conical Branch	Jones 1969, Fig. 22; UMC 1986, Report SRF386

Code	Official Fitting Title / Description	System	Shape	Category	Research Citation
SD5-17	Tee, Conical Branch with 60 Deg. Elbow, Branch 45 Deg. to Main, Diverging	Supply	Round	Conical Branch	Jones 1969, Fig. 23; UMC 1986, Report SRF386
SD5-18	Bullhead Tee without Vanes, Diverging	Supply	Round	Bullhead	Sepsy 1969
SD5-19	Bullhead Tee with Vanes, Diverging	Supply	Round	Bullhead	Sepsy 1969
SD5-2	Wye, 45 Degree, Conical Branch Tapered into Body, Diverging	Supply	Round	Conical Branch	UMC 1986, Report SRF386
SD5-20	Capped Cross, 90 Degree, Diverging	Supply	Round	Cross	Sepsy 1969
SD5-21	Heel-Tapped Elbow, 90 Degree, r/Dc = 1.5, Diverging	Supply	Round	Junctions, Diverging	UMC 1990
SD5-22	Symmetrical Wye, w/45 Deg. Elbows	Supply	Round	Non-Conical Branch	Branches 90 Degrees to Main, Diverging (Sepsy 1969
SD5-23	Double Wye, 45 Degree, Diverging	Supply	Round	Junctions, Diverging	UMC 1990, Fig. 33; UMC 1986, Report SRF386
SD5-24	Cross, Diverging	Supply	Round	Cross	UMC 1990, Fig. 33; UMC 1986, Report SRF386
SD5-25	Cross, Conical Branches Tapered into Body, Diverging	Supply	Round	Cross	UMC 1990, Fig. 33; UMC 1986, Report SRF386
SD5-26	Cross, 45 Degree Entry Branches, Diverging	Supply	Round	Cross	UMC 1990, Fig. 33; UMC 1986, Report SRF386
SD5-27	Symetrical Wye, 90 Degrees, Diverging	Supply	Round	Conical Branch	Sepsy 1969
SD5-3	Wye, 45 Degree, Straight Body Branch with Reducing Transition, Diverging	Supply	Round	Non-Conical Branch	Jones 1969, Fig. 14: UMC 1986, Report SRF386
SD5-4	Wye, 45 Degree, Branch with 45 Degree Elbow, Branch 90 Degrees to Main	Supply	Round	Non-Conical Branch	UMC 1986, Report SRF386
SD5-5	Wye, 45 Deg., Conical Br.with 60 Deg. Elbow, Br. 90 Deg. to	Supply	Round	Conical Branch	Main, Diverging (Jones 1969, Fig. 20; UMC 1986, Report SRF386
SD5-6	Wye, 45 Degree, Branch with 60 Degree Elbow, Branch 90 Degrees	Supply	Round	Non-Conical Branch	to Main, Diverging (Jones 1963, Fig. 3; UMC 1986, Report SRF386
SD5-7	Wye, 45 Degree, Branch with 30 Deg. Elbow, Branch 45 Deg. to	Supply	Round	Non-Conical Branch	Main, Diverging (Jones 1963, Fig. 2; UMC 1986, Report SRF386
SD5-8	Wye, 45 Deg., Conical Br. with 30 Deg. Elbow, Br. 45 Deg. to	Supply	Round	Conical Branch	Main, Diverging (Jones 1969, Fig. 24; UMC 1986, Report SRF386
SD5-9	Tee, Diverging	Supply	Round	Non-Conical Branch	UMC 1986, Report SRF386
SD7-1	Fan, Vaneaxial, Discharging into Plenum	Supply	Round	Fan & System Interactions	AMCA 1990, Fig. 8-2
SD7-2	Fan Outlet, Tubeaxial, with 4-Gore Elbow	Supply	Round	Fan & System Interactions	AMCA 1990, Fig. 8-4
SD7-3	Fan Outlet, Tubeaxial, with Mitered Elbow	Supply	Round	Fan & System Interactions	AMCA 1990, Fig. 8-4
SD7-4	Fan Outlet, Vaneaxial, with 4-Gore Elbow	Supply	Round	Fan & System Interactions	AMCA 1990, Fig. 8-4
SD7-5	Fan Outlet, Vaneaxial, with Mitered Elbow	Supply	Round	Fan & System Interactions	AMCA 1990, Fig. 8-4
SF4-1	Transition, Flat Oval, Concentric && Eccentric, Supply Air Systems	Supply	Flat Oval	Transitions	Zhang, Sun, Ford && Denny 2014
SF5-1	Tee, Flat Oval, Straight Body Branch, Diverging	Supply	Flat Oval	Junctions, Diverging	Gibbs && Idem 2012
SF5-2	Tee, Flat Oval, Conical Body Branch, Diverging	Supply	Flat Oval	Junctions, Diverging	Townsend et al. 1996; Idem && Khodabakhsh 1999
SF5-3	Wye, Flat Oval, 45 Degree, Straight Body Branch, Diverging	Supply	Flat Oval	Junctions, Diverging	Gibbs && Idem 2012
SF5-4	Wye, Flat Oval, 45 Degree, Conical Body Branch, Diverging	Supply	Flat Oval	Junctions, Diverging	Townsend et al. 1996; Idem && Khodabakhsh 1999
SR1-1	Conical Bellmouth/Sudden Contraction, Plenum to	Supply	Rectangular	Plenums	Rectangular, Supply Air Systems (Idelchik 1986, Diagram 4-9 and 3-7
SR2-1	Abrupt Exit	Supply	Rectangular	Exits	Idelchik 1986, Diagram 11-1
SR2-2	Discharge from Duct Flush with Wall	Supply	Rectangular	Exits	Idelchik 1986, Diagram 11-2
SR2-3	Plain Diffuser (Two Sides Parallel), Free Discharge	Supply	Rectangular	Exits	Idelchik 1986, Diagram 11-6
SR2-4	Plain Diffuser (Two Sides Parallel), with Wall	Supply	Rectangular	Exits	Idelchik 1986, Diagram 5-10
SR2-5	Pyramidal Diffuser, Free Discharge	Supply	Rectangular	Exits	Idelchik 1986, Diagram 11-5
SR2-6	Pyramidal Diffuser, with Wall	Supply	Rectangular	Exits	Idelchik 1986, Diagram 5-9
SR3-1	Elbow, 90 Degree, Variable Inlet/Outlet Areas, Supply Air Systems	Supply	Rectangular	Elbows	Idelchik 1986, Diagram 6-4
SR4-1	Transition, Rectangular, Two Sides Parallel, Symmetrical, Supply	Supply	Rectangular	Transitions	Air Systems (Idelchik 1986, Diagram 5-5
SR4-2	Transition, Pyramidal, Supply Air Systems	Supply	Rectangular	Transitions	Idelchik 1986, Diagram 5-4
SR4-3	Transition, Round to Rectangular, Supply Air Systems	Supply	Rectangular	Transitions	Idelchik 1986, Diagram 5-27
SR5-1	Smooth Radius Wye of the Type As + Ab > or = Ac, Branch	Supply	Rectangular	Wye	90 Degrees to Main, Diverging (Idelchik 1986, Diagram 7-21
SR5-10	Tee of the Type As + Ab = Ac, Diverging	Supply	Rectangular	Tee	Idelchik 1986, Diagrams 7-16 && 7-17
SR5-11	Tee, Rectangular Main to Round Tap, Diverging (Brooks 1993)	Supply	Rectangular	Tee	

Code	Official Fitting Title / Description	System	Shape	Category	Research Citation
SR5-12	Tee, Rectangular Main to Round Conical Tap, Diverging	Supply	Rectangular	Tee	Inoue 1980; Korst 1950; Brooks 1993
SR5-13	Tee, 45 Degree Entry Branch, Diverging (Brooks 1993)	Supply	Rectangular	Tee	
SR5-14	Wye, Symmetrical, Dovetail, Qb/Qc = 0.5, Diverging	Supply	Rectangular	Symmetrical	Idelchik 1986, Diagram 7-24
SR5-15	Bullhead Tee without Vanes, Diverging	Supply	Rectangular	Tee	Idelchik 1986, Diagrams 7-16 && 7-30
SR5-20	Double Wye of the Type As = Ac, 45 Degree, Diverging	Supply	Rectangular	Junctions, Diverging	Idelchik 1986, Diagrams 7-15, 7-17 && 7-27
SR5-21	Cross of the Type As = Ac, Diverging	Supply	Rectangular	Junctions, Diverging	Idelchik 1986, Diagrams 7-15, 7-17 && 7-29
SR5-3	Wye of the Type As + Ab > Ac, As = Ac, 45 Degree, Diverging	Supply	Rectangular	Unsymmetrical	Idelchik 1986, Diagrams 7-15 && 7-17
SR5-5	Tee of the Type As + Ab > Ac, As = Ac, Diverging	Supply	Rectangular	Tee	Idelchik 1986, Diagrams 7-15 && 7-17
SR5-8	Wye of the Type As + Ab = Ac, 45 Degree, Diverging	Supply	Rectangular	Unsymmetrical	Idelchik 1986, Diagrams 7-16 && 7-17
SR7-1	Fan, Centrifugal, without an Outlet Diffuser, Free Discharge	Supply	Rectangular	Fan & System Interactions	AMCA 1990, Fig. 8-3
SR7-10	Fan Outlet, Centrifugal, DWDI, with Elbow (Position B)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5
SR7-11	Fan Outlet, Centrifugal, DWDI, with Elbow (Position C)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5
SR7-12	Fan Outlet, Centrifugal, DWDI, with Elbow (Position D)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5
SR7-13	Plane Symmetric Diffuser at Centrifugal Fan Outlet with	Supply	Rectangular	Duffuser at Centrifugal Fan Outlet	Ductwork (Idelchik 1986, Diagram 5-12
SR7-14	Plane Asymmetric Diffuser (Configuration 1) at Centrifugal	Supply	Rectangular	Plane Asymmetric	Fan Outlet with Ductwork (Idelchik 1986, Diagram 5-13
SR7-15	Plane Asymmetric Diffuser (Configuration 2) at Centrifugal	Supply	Rectangular	Plane Asymmetric	Fan Outlet with Ductwork (Idelchik 1986, Diagram 5-14
SR7-16	Plane Asymmetric Diffuser (Configuration 3) at Centrifugal	Supply	Rectangular	Plane Asymmetric	Fan Outlet with Ductwork (Idelchik 1986, Diagram 5-15
SR7-17	Pyramidal Diffuser at Centrifugal Fan Outlet with Ductwork	Supply	Rectangular	Duffuser at Centrifugal Fan Outlet	Idelchik 1986, Diagram 5-16
SR7-2	Plane Asymmetric Diffuser at Centrifugal Fan Outlet	Supply	Rectangular	Plane Asymmetric	Free Discharge (Idelchik 1986, Diagram 11-11
SR7-3	Pyramidal Diffuser at Centrifugal Fan Outlet, Free Discharge	Supply	Rectangular	Duffuser at Centrifugal Fan Outlet	Idelchik 1986, Diagram 11-11
SR7-4	Fan, Centrifugal, Discharging into a Plenum	Supply	Rectangular	Fan & System Interactions	AMCA 1990, Fig. 8-3
SR7-5	Fan Outlet, Centrifugal, SWSI, with Elbow (Position A)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5
SR7-6	Fan Outlet, Centrifugal, SWSI, with Elbow (Position B)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5
SR7-7	Fan Outlet, Centrifugal, SWSI, with Elbow (Position C)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5
SR7-8	Fan Outlet, Centrifugal, SWSI, with Elbow (Position D)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5
SR7-9	Fan Outlet, Centrifugal, DWDI, with Elbow (Position A)	Supply	Rectangular	Fan Outlet, Centrifugal	AMCA 1990, Fig. 8-5

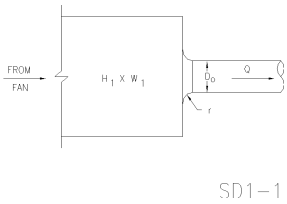
SUPPLY AIR SYSTEMS — EXPANDED CATALOG

Supply • Round Duct Catalog (43 Fittings)

SD1-1 Bellmouth (Standard Radius), Plenum to Round, Supply Air Systems (Idelchik 1986, Diagram 4-9 and 3-4)

ROUND • PLENUMS

Hierarchy Path: Supply > Round > Plenums > **SD1-1**



SD1-1

Var	Parameter	SI Def	IP Def	Range
Do	Diameter	300	12	100–1200
H1	Plenum Height	6250	250	150–12500
W1	Plenum Width	6250	250	150–12500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

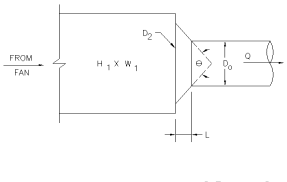
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
r	Bellmouth Radius	(r, in.)	(r, in.)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Po, Pa)	(in. wg)	0

Governing Loss Formulation: `xCo=fcn((r/Do),2) ; Co=(1-Ao/A1)^0.75*xCo ; Po=Co*Pvo`

SD1-2 Conical Bellmouth/Sudden Contraction, Plenum to Round (Supply Air Systems (Idelchik 1986, Diagram 4-9)

ROUND • PLENUMS

Hierarchy Path: Supply > Round > Plenums > **SD1-2**



SD1-2

Var	Parameter	SI Def	IP Def	Range
Do	Diameter	600	24	75–3000
H1	Plenum Height	6250	250	150–12500
W1	Plenum Width	6250	250	150–12500
Theta	Angle	60	60	0–180
L	Length	175	7	0–1500
Q	Flow Rate	2400	5000	5–9999999
rho	Density	1.204	0.075	> 0

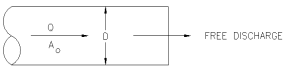
Var	Output Metric	SI Unit	IP Unit	Dec
XL	Length	(L, mm)	(L, in.)	0
Theta 2	Angle	(Theta, deg.)	(Theta, deg.)	0
D2	Diameter	(D2, mm)	(D2, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Po, Pa)	(in. wg)	0

Governing Loss Formulation: `xCo=fcn(Theta2,XL/Do,1) ; Co=(1-Ao/A1)^0.75*xCo ; Po=Co*Pvo`

SD2-1 Abrupt Exit (Idelchik 1986, Diagram 11-1)

ROUND • EXITS

Hierarchy Path: Supply > Round > Exits > **SD2-1**



Var	Parameter	SI Def	IP Def	Range
D	Diameter	300	12	75–3000
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

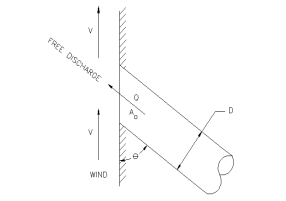
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=IFLE0(Re-3000.,2.0,1.0) ; Po=Co*Pv`

SD2-2 Discharge from a Duct Flush with Wall (Idelchik 1986, Diagram 11-2)

ROUND • EXITS

Hierarchy Path: Supply > Round > Exits > **SD2-2**



Var	Parameter	SI Def	IP Def	Range
D	Diameter	300	12	75–3000
Theta	Angle	90	90	30–150
V	Velocity	0.5	100	0–25
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

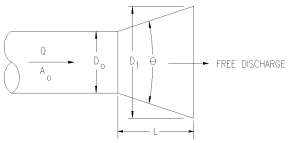
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((V/Vo),Theta,1) ; Po=Co*Pv`

SD2-3 Conical Diffuser, Free Discharge (Idelchik 1986, Diagram 11-3)

ROUND • EXITS

Hierarchy Path: Supply > Round > Exits > **SD2-3**



SD2-3

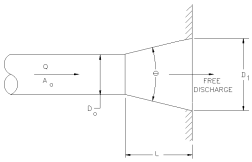
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	75–3000	**D1**	Diameter	(D1, mm)	(D1, in.)	0
Theta	Angle	45	45	3–120	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	450	18	0–1200	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,Re,(A1/Ao),1) ; Po=Co*Pv`

SD2-4 Conical Diffuser, with Wall (Idelchik 1986, Diagram 5-8)

ROUND • EXITS

Hierarchy Path: Supply > Round > Exits > **SD2-4**



SD2-4

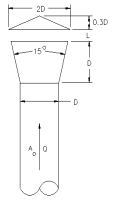
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	75–3000	**Theta**	Optimum Angle	(Theta, deg.)	(Theta, deg.)	0
L	Length	450	18	25–1500	**D1**	Diameter	(D1, mm)	(D1, in.)	0
Q	Flow Rate	700	1500	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(L/Do,1) ; Po=Co*Pv`

SD2-5 Exhaust Hood (Idelchik 1986, Diagram 11-16)

ROUND • EXITS

Hierarchy Path: Supply > Round > Exits > **SD2-5**



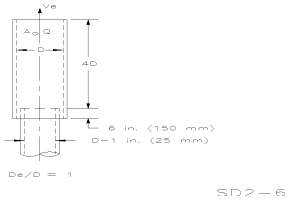
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	150	6	25–1500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(L/D,1) ; Po=Co*Pv`

SD2-6 Stackhead (Dexit/D=1) (Idelchik 1986, Diagram 11-23)

ROUND • EXITS

Hierarchy Path: Supply > Round > Exits > **SD2-6**



SD2-6

Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Stack Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Vo**	Exit Velocity	(Ve, m/s)	(Ve, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(De/D,1) ; Po=Co*Pv`

SD2-7 Stackhead (Dexit/D<1) (Idelchik 1986, Diagram 11-23)

ROUND • EXITS

Hierarchy Path: Supply > Round > Exits > **SD2-7**



SD2-7

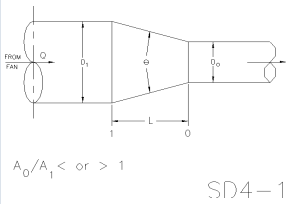
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Stack Velocity	(Vo, m/s)	(Vo, fpm)	1
De	Diameter	200	8	75–2500	**Ve**	Exit Velocity	(Ve, m/s)	(Ve, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(De/D,1) ; Po=Co*Pv`

SD4-1 Transition, Round to Round, Supply Air Systems (Idelchik 1986, Diagram 5-2 && 5-22)

ROUND • TRANSITIONS

Hierarchy Path: Supply > Round > Transitions > **SD4-1**



Var	Parameter	SI Def	IP Def	Range
Do	Diameter	300	12	75–3000
D1	Diameter	200	8	75–3000
L	Length	300	12	0–1500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

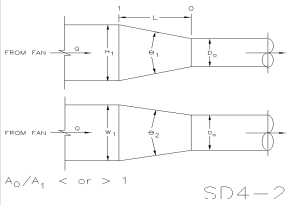
Var	Output Metric	SI Unit	IP Unit	Dec
Theta	Angle	(Theta, deg.)	(Theta, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[Theta,(Ao/A1),1] ; Po=Co*Pvo`

SD4-2 Transition, Rectangular to Round, Supply Air Systems (Idelchik 1986, Diagram 5-27)

ROUND • TRANSITIONS

Hierarchy Path: Supply > Round > Transitions > **SD4-2**



Var	Parameter	SI Def	IP Def	Range
H1	Height	600	24	75–4000
W1	Width	600	24	75–4000
Do	Diameter	800	32	75–3000
L	Length	450	18	0–1500
Q	Flow Rate	4000	8000	5–9999999
rho	Density	1.204	0.075	> 0

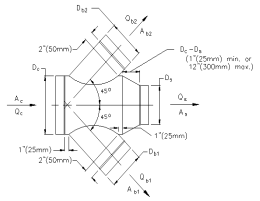
Var	Output Metric	SI Unit	IP Unit	Dec
Theta 1	Angle	(Theta1, deg.)	(Theta1, deg.)	0
Theta 2	Angle	(Theta2, deg.)	(Theta2, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[Theta,Ao/A1,1] ; Po=Co*Pvo`

SD5-23 Double Wye, 45 Degree, Diverging (UMC 1990, Fig. 33; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > **SD5-23**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Ds	Diameter	275	11	50–2500
Db1	Diameter	200	8	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	250	500	5–9999999
Qb2	Flow Rate	250	500	5–9999999
rho	Density	1.204	0.075	> 0

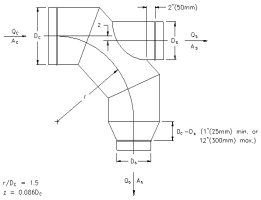
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn[(Qs/Qc),(As/Ac),2] ; Pos=Cs*Pvs`

SD5-21 Heel-Tapped Elbow, 90 Degree, r/Dc = 1.5, Diverging (UMC 1990)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > **SD5-21**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

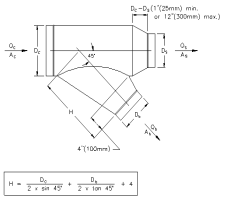
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-2 Wye, 45 Degree, Conical Branch Tapered into Body, Diverging (UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Conical Branch > **SD5-2**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

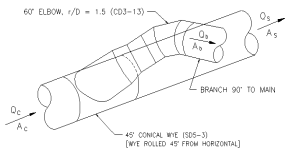
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-5 Wye, 45 Deg., Conical Br.with 60 Deg. Elbow, Br. 90 Deg. to (Main, Diverging (Jones 1969, Fig. 20; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Conical Branch > **SD5-5**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

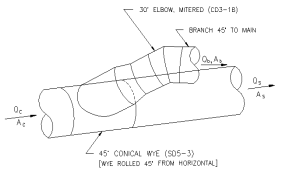
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-8 Wye, 45 Deg., Conical Br. with 30 Deg. Elbow, Br. 45 Deg. to (Main, Diverging (Jones 1969, Fig. 24; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Conical Branch > **SD5-8**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

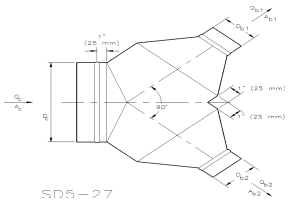
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-27 Symetrical Wye, 90 Degrees, Diverging (Sepsy 1969)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Conical Branch > **SD5-27**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db1	Diameter	150	6	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

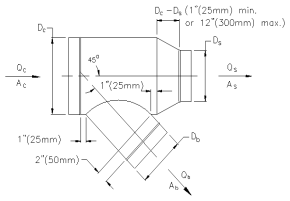
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=(PI/4)*(Dc/ADC)^2 ; Ab1=(PI/4)*(Db1/ADC)^2 ; Ab2=(PI/4)*(Db2/ADC)^2`

SD5-1 Wye, 45 Degree, Diverging (UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Non-Conical Branch > **SD5-1**



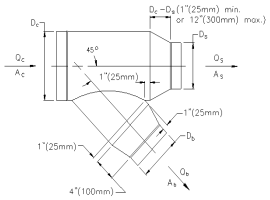
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-3 Wye, 45 Degree, Straight Body Branch with Reducing Transition, Diverging (Jones 1969, Fig. 14: UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Non-Conical Branch > **SD5-3**



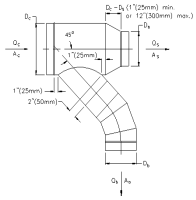
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-4 Wye, 45 Degree, Branch with 45 Degree Elbow, Branch 90 Degrees to Main (UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Non-Conical Branch > **SD5-4**



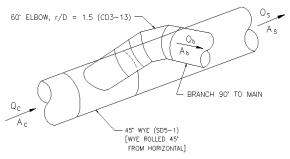
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	Vb, fpm)	1
Db	Diameter	150	6	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-6 Wye, 45 Degree, Branch with 60 Degree Elbow, Branch 90 Degrees (to Main, Diverging (Jones 1963, Fig. 3; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Non-Conical Branch > **SD5-6**



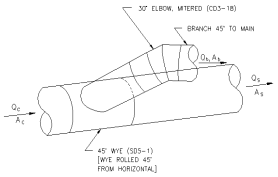
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-7 Wye, 45 Degree, Branch with 30 Deg. Elbow, Branch 45 Deg. to (Main, Diverging (Jones 1963, Fig. 2; UMC 1986, Report SRF386))

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Non-Conical Branch > **SD5-7**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

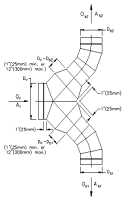
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-22 Symmetrical Wye, w/45 Deg. Elbows (Branches 90 Degrees to Main, Diverging (Sepsy 1969))

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Wye > Non-Conical Branch > **SD5-22**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db1	Diameter	150	6	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

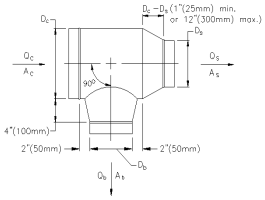
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=(PI/4)*(Dc/ADC)^2 ; Ab1=(PI/4)*(Db1/ADC)^2 ; Ab2=(PI/4)*(Db2/ADC)^2`

SD5-10 Tee, Conical Branch Tapered into Body, Diverging (UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Conical Branch > **SD5-10**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

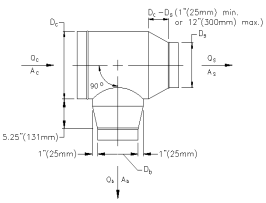
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-11 Tee, Conical Branch, Diverging (Jones 1969, Fig. 12; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Conical Branch > **SD5-11**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

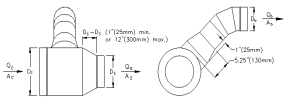
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-14 Tee, Conical Branch with 45 Deg. Elbow, Branch 90 Deg. to Main, Diverging (Jones 1969, Fig. 19; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Conical Branch > **SD5-14**



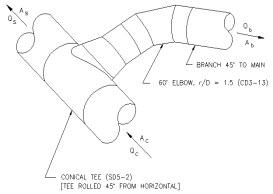
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	Cb	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	Pob	Branch Pressure Loss	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	Main Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-17 Tee, Conical Branch with 60 Deg. Elbow, Branch 45 Deg. to Main, Diverging (Jones 1969, Fig. 23; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Conical Branch > **SD5-17**



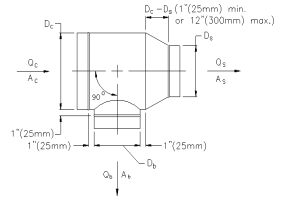
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	Cb	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	Pob	Branch Pressure Loss	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	Main Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-9 Tee, Diverging (UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Non-Conical Branch > **SD5-9**



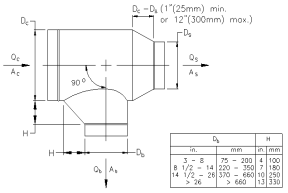
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	Cb	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	Pob	Branch Pressure Loss	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	Main Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-12 Tee, 45 Degree Entry Branch, Diverging (UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Non-Conical Branch > **SD5-12**



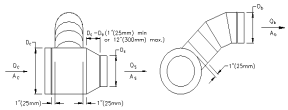
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Dc	Diameter	300	12	50–2500	Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Ds	Diameter	250	10	50–2500	Cb	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	Pob	Branch Pressure Loss	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	Main Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-13 Tee, Branch with 45 Degree Elbow, Branch 90 Degrees to Main, Diverging (Jones 1969, Fig. 18; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Non-Conical Branch > **SD5-13**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

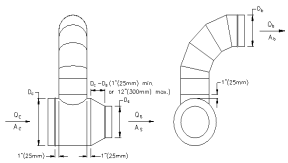
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-15 Tee, Branch with 90 Degree Elbow, Branch 90 Degrees to Main, Diverging (Jones 1969, Fig. 17; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Non-Conical Branch > **SD5-15**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

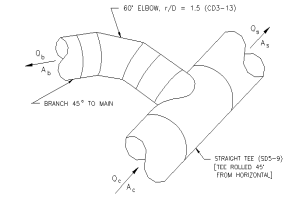
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-16 Tee, Branch with 60 Degree Elbow, Branch 45 Degrees to Main, Diverging (Jones 1969, Fig. 22; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Non-Conical Branch > **SD5-16**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db	Diameter	150	6	25–2500
Ds	Diameter	250	10	50–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

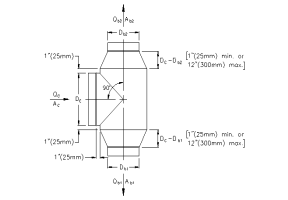
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),(Ab/Ac),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),(As/Ac),2]`

SD5-18 Bullhead Tee without Vanes, Diverging (Sepsy 1969)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Bullhead > **SD5-18**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db1	Diameter	150	6	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

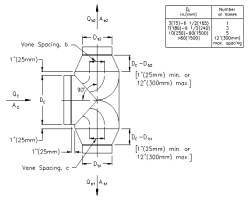
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=(PI/4)*(Dc/ADC)^2 ; Ab1=(PI/4)*(Db1/ADC)^2 ; Ab2=(PI/4)*(Db2/ADC)^2`

SD5-19 Bullhead Tee with Vanes, Diverging (Sepsy 1969)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Tee > Bullhead > **SD5-19**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db1	Diameter	150	6	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

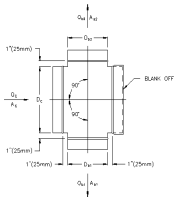
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=(PI/4)*(Dc/ADC)^2 ; Ab1=(PI/4)*(Db1/ADC)^2 ; Ab2=(PI/4)*(Db2/ADC)^2`

SD5-20 Capped Cross, 90 Degree, Diverging (Sepsy 1969)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Cross > **SD5-20**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Db1	Diameter	150	6	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

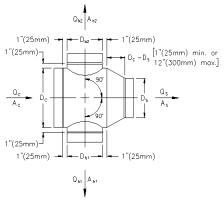
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=(PI/4)*(Dc/ADC)^2 ; Ab1=(PI/4)*(Db1/ADC)^2 ; Ab2=(PI/4)*(Db2/ADC)^2`

SD5-24 Cross, Diverging (UMC 1990, Fig. 33; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Cross > **SD5-24**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Ds	Diameter	275	11	50–2500
Db1	Diameter	200	8	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	250	500	5–9999999
Qb2	Flow Rate	250	500	5–9999999
rho	Density	1.204	0.075	> 0

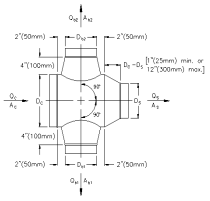
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn[(Qs/Qc),(As/Ac),2] ; Pos=Cs*Pvs`

SD5-25 Cross, Conical Branches Tapered into Body, Diverging (UMC 1990, Fig. 33; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Cross > **SD5-25**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Ds	Diameter	275	11	50–2500
Db1	Diameter	200	8	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	250	500	5–9999999
Qb2	Flow Rate	250	500	5–9999999
rho	Density	1.204	0.075	> 0

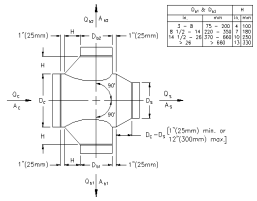
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn[(Qs/Qc),(As/Ac),2] ; Pos=Cs*Pvs`

SD5-26 Cross, 45 Degree Entry Branches, Diverging (UMC 1990, Fig. 33; UMC 1986, Report SRF386)

ROUND • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Round > Junctions, Diverging > Cross > **SD5-26**



Var	Parameter	SI Def	IP Def	Range
Dc	Diameter	300	12	50–2500
Ds	Diameter	275	11	50–2500
Db1	Diameter	200	8	25–2500
Db2	Diameter	200	8	25–2500
Qc	Flow Rate	1000	2000	5–9999999
Qb1	Flow Rate	250	500	5–9999999
Qb2	Flow Rate	250	500	5–9999999
rho	Density	1.204	0.075	> 0

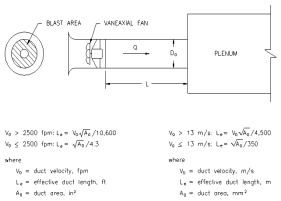
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn[(Qs/Qc),(As/Ac),2] ; Pos=Cs*Pvs`

SD7-1 Fan, Vaneaxial, Discharging into Plenum (AMCA 1990, Fig. 8-2)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Round > Fan & System Interactions > **SD7-1**



Var	Parameter	SI Def	IP Def	Range
Do	Diameter	300	12	300–3000
L	Length	300	12	0–6000
Q	Flow Rate	1180	2500	5–9999999
rho	Density	1.204	0.075	> 0

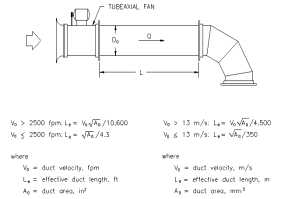
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(LOLe,1) ; Po=Co*Pv`

SD7-2 Fan Outlet, Tubeaxial, with 4-Gore Elbow (AMCA 1990, Fig. 8-4)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Round > Fan & System Interactions > **SD7-2**



Var	Parameter	SI Def	IP Def	Range
Do	Diameter	300	12	300–3000
L	Length	600	24	0–6000
Q	Flow Rate	1250	2500	5–9999999
rho	Density	1.204	0.075	> 0

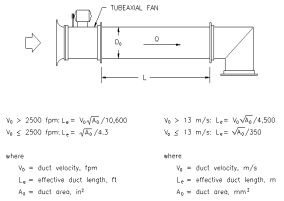
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.0 ; Po=Co*Pv`

SD7-3 Fan Outlet, Tubeaxial, with Mitered Elbow (AMCA 1990, Fig. 8-4)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Round > Fan & System Interactions > **SD7-3**



$V_0 > 2500$ fpm: $L_e = V_0\sqrt{A_0}/10{,}600$
$V_0 \le 2500$ fpm: $L_e = \sqrt{A_0}/4.3$

where
V_0 = duct velocity, fpm
L_e = effective duct length, ft
A_0 = duct area, in²

$V_0 > 13$ m/s: $L_e = V_0\sqrt{A_0}/4{,}500$
$V_0 \le 13$ m/s: $L_e = \sqrt{A_0}/350$

where
V_0 = duct velocity, m/s
L_e = effective duct length, m
A_0 = duct area, mm²

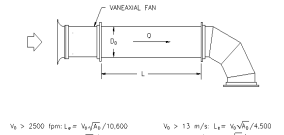
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	600	24	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	1250	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.0 ; Po=Co*Pv`

SD7-4 Fan Outlet, Vaneaxial, with 4-Gore Elbow (AMCA 1990, Fig. 8-4)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Round > Fan & System Interactions > **SD7-4**



$V_0 > 2500$ fpm: $L_e = V_0\sqrt{A_0}/10{,}600$
$V_0 \le 2500$ fpm: $L_e = \sqrt{A_0}/4.3$

where
V_0 = duct velocity, fpm
L_e = effective duct length, ft
A_0 = duct area, in²

$V_0 > 13$ m/s: $L_e = V_0\sqrt{A_0}/4{,}500$
$V_0 \le 13$ m/s: $L_e = \sqrt{A_0}/350$

where
V_0 = duct velocity, m/s
L_e = effective duct length, m
A_0 = duct area, mm²

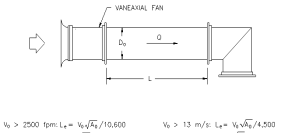
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	0	0	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	1180	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(L0Le,1) ; Po=Co*Pv`

SD7-5 Fan Outlet, Vaneaxial, with Mitered Elbow (AMCA 1990, Fig. 8-4)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Round > Fan & System Interactions > **SD7-5**



$V_0 > 2500$ fpm: $L_e = V_0\sqrt{A_0}/10{,}600$
$V_0 \le 2500$ fpm: $L_e = \sqrt{A_0}/4.3$

where
V_0 = duct velocity, fpm
L_e = effective duct length, ft
A_0 = duct area, in²

$V_0 > 13$ m/s: $L_e = V_0\sqrt{A_0}/4{,}500$
$V_0 \le 13$ m/s: $L_e = \sqrt{A_0}/350$

where
V_0 = duct velocity, m/s
L_e = effective duct length, m
A_0 = duct area, mm²

Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	300	12	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	1180	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

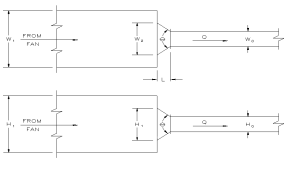
Governing Loss Formulation: `Co=fcn(L0Le,1) ; Po=Co*Pv`

Supply • Rectangular Duct Catalog (40 Fittings)

SR1-1 Conical Bellmouth/Sudden Contraction, Plenum to (Rectangular, Supply Air Systems (Idelchik 1986, Diagram 4-9 and 3-7)

RECTANGULAR • PLENUMS

Hierarchy Path: Supply > Rectangular > Plenums > **SR1-1**



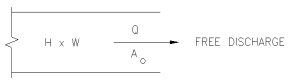
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	600	24	75–4000	**XL**	Length	(L, mm)	(L, in.)	0
Wo	Width	600	24	75–4000	**Theta 2**	Angle	(Theta, deg.)	(Theta, deg.)	0
H1	Plenum Height	6250	250	150–12500	**H2**	Length	(H2, mm)	(H2, in.)	0
W1	Plenum Width	6250	250	150–12500	**W2**	Length	(W2, mm)	(W2, in.)	0
Theta	Angle	30	30	0–180	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	175	7	0–1200	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
Q	Flow Rate	2100	4500	5–9999999	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
rho	Density	1.204	0.075	> 0	**Pv1**	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `xCo=fcn(Theta2,XL/Dh,1) ; Co=(1-Ao/A1)^0.75*xCo ; Po=Co*Pvo`

SR2-1 Abrupt Exit (Idelchik 1986, Diagram 11-1)

RECTANGULAR • EXITS

Hierarchy Path: Supply > Rectangular > Exits > **SR2-1**



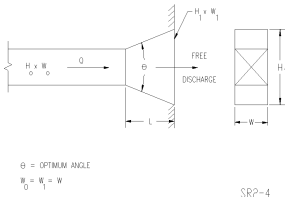
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
H	Height	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=IFLE0((Re-3000.),Hw,1.0) ; Po=Co*Pv`

SR2-4 Plain Diffuser (Two Sides Parallel), with Wall (Idelchik 1986, Diagram 5-10)

RECTANGULAR • EXITS

Hierarchy Path: Supply > Rectangular > Exits > **SR2-4**



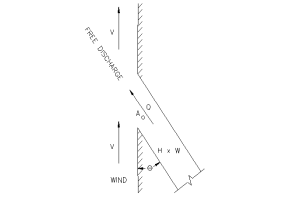
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	300	12	75–4000	**Theta**	Optimum Angle	(Theta, deg.)	(Theta, deg.)	0
Wo	Width	300	12	75–4000	**H1**	Height	(H1, mm)	(H1, in.)	0
L	Length	250	10	25–1500	**W1**	Width	(W1, mm)	(W1, in.)	0
Q	Flow Rate	1000	2000	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(L/Ho,1) ; Po=Co*Pv`

SR2-2 Discharge from Duct Flush with Wall (Idelchik 1986, Diagram 11-2)

RECTANGULAR • EXITS

Hierarchy Path: Supply > Rectangular > Exits > **SR2-2**



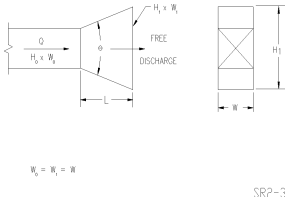
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
H	Height	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Theta	Angle	90	90	30–150	**Co**	Loss Coefficient	(Co)	(Co)	2
V	Velocity	0.5	100	0–25	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	700	1500	5–9999999					
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn[(V/Vo),Theta,(H/W),1] ; Po=Co*Pv`

SR2-3 Plain Diffuser (Two Sides Parallel), Free Discharge (Idelchik 1986, Diagram 11-6)

RECTANGULAR • EXITS

Hierarchy Path: Supply > Rectangular > Exits > **SR2-3**



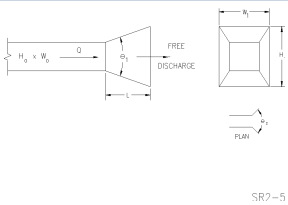
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	300	12	75–4000	**H1**	Height	(H1, mm)	(H1, in.)	0
Wo	Width	300	12	75–4000	**W1**	Width	(W1, mm)	(W1, in.)	0
Theta	Angle	45	45	4–120	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	250	10	25–1200	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	1000	2000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[Theta,Re,(A1/Ao),1] ; Po=Co*Pv`

SR2-5 Pyramidal Diffuser, Free Discharge (Idelchik 1986, Diagram 11-5)

RECTANGULAR • EXITS

Hierarchy Path: Supply > Rectangular > Exits > **SR2-5**



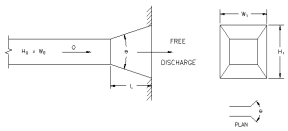
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	300	12	75–4000	**H1**	Height	(H1, mm)	(H1, in.)	0
Wo	Width	450	18	75–4000	**W1**	Width	(W1, mm)	(W1, in.)	0
Theta	Angle	45	45	0–120	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	250	10	25–2500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[Theta,Re,(A1/Ao),1] ; Po=Co*Pv`

SR2-6 Pyramidal Diffuser, with Wall (Idelchik 1986, Diagram 5-9)

RECTANGULAR • EXITS

Hierarchy Path: Supply > Rectangular > Exits > **SR2-6**



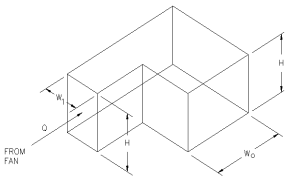
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	300	12	75–4000	**Theta**	Optimum Angle	(Theta, deg.)	(Theta, deg.)	0
Wo	Width	300	12	75–4000	**H1**	Height	(H1, mm)	(H1, in.)	0
L	Length	450	18	25–1500	**W1**	Width	(W1, mm)	(W1, in.)	0
Q	Flow Rate	700	1500	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(L/Dh,1) ; Po=Co*Pv`

SR3-1 Elbow, 90 Degree, Variable Inlet/Outlet Areas, Supply Air Systems (Idelchik 1986, Diagram 6-4)

RECTANGULAR • ELBOWS

Hierarchy Path: Supply > Rectangular > Elbows > **SR3-1**



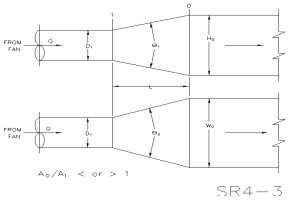
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	250	10	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Wo	Width	500	20	75–4000	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
W1	Width	250	10	75–4000	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Pv1**	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[(Wo/W1),(H/W1),1] ; Po=Co*Pvo`

SR4-3 Transition, Round to Rectangular, Supply Air Systems (Idelchik 1986, Diagram 5-27)

RECTANGULAR • TRANSITIONS

Hierarchy Path: Supply > Rectangular > Transitions > **SR4-3**



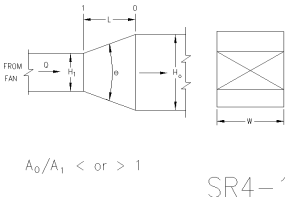
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D1	Diameter	300	12	75–3000	**Theta 1**	Angle	(Theta1, deg.)	(Theta1, deg.)	0
Ho	Height	600	24	75–4000	**Theta 2**	Angle	(Theta2, deg.)	(Theta2, deg.)	0
Wo	Width	600	24	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	600	24	0–6000	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
rho	Density	1.204	0.075	> 0	**Pv1**	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					C1	Loss Coefficient	(C1)	(C1)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,(Ao/A1),1) ; Po=Co*Pvo`

SR4-1 Transition, Rectangular, Two Sides Parallel, Symmetrical, Supply (Air Systems (Idelchik 1986, Diagram 5-5)

RECTANGULAR • TRANSITIONS

Hierarchy Path: Supply > Rectangular > Transitions > **SR4-1**



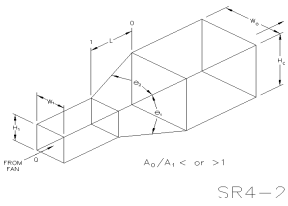
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	300	12	75–4000	**Theta**	Angle	(Theta, deg.)	(Theta, deg.)	0
W	Width	400	16	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
H1	Height	600	24	75–4000	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
L	Length	600	24	0–2500	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Q	Flow Rate	1650	3500	5–9999999	**Pv1**	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					C1	Loss Coefficient	(C1)	(C1)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[Theta,(Ao/A1),1] ; Po=Co*Pvo`

SR4-2 Transition, Pyramidal, Supply Air Systems (Idelchik 1986, Diagram 5-4)

RECTANGULAR • TRANSITIONS

Hierarchy Path: Supply > Rectangular > Transitions > **SR4-2**



Var	Parameter	SI Def	IP Def	Range
Ho	Height	450	18	75–4000
Wo	Width	450	18	75–4000
H1	Height	300	12	75–4000
W1	Width	300	12	75–4000
L	Length	600	24	0–2500
Q	Flow Rate	1000	2000	5–9999999
rho	Density	1.204	0.075	> 0

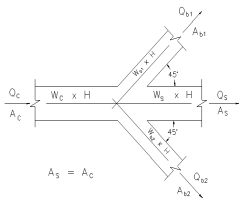
Var	Output Metric	SI Unit	IP Unit	Dec
Theta 1	Angle	(Theta1, deg.)	(Theta1, deg.)	0
Theta 2	Angle	(Theta2, deg.)	(Theta2, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[Theta,(Ao/A1),1] ; Po=Co*Pvo`

SR5-20 Double Wye of the Type As = Ac, 45 Degree, Diverging (Idelchik 1986, Diagrams 7-15, 7-17 && 7-27)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > **SR5-20**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Wb1	Width	200	8	25–2500
Wb2	Width	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	400	300	5–9999999
Qb2	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

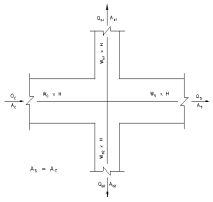
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn((Qs/Qc),2) ; Pos=Cs*Pvs`

SR5-21 Cross of the Type As = Ac, Diverging (Idelchik 1986, Diagrams 7-15, 7-17 && 7-29)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > **SR5-21**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Wb1	Width	200	8	25–2500
Wb2	Width	200	8	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb1	Flow Rate	400	800	5–9999999
Qb2	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

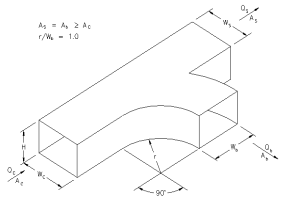
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn((Qs/Qc),2) ; Pos=Cs*Pvs`

SR5-1 Smooth Radius Wye of the Type As + Ab > or = Ac, Branch (90 Degrees to Main, Diverging (Idelchik 1986, Diagram 7-21))

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Wye > **SR5-1**



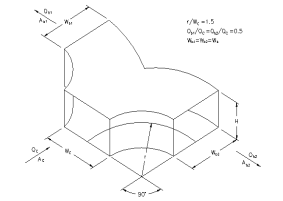
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Wc	Width	300	12	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Wb	Width	150	6	25–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Ws	Width	250	10	25–2500	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qc	Flow Rate	750	1500	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Qb	Flow Rate	150	300	5–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),(As/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(Ab/Ac),(As/Ac),2)`

SR5-14 Wye, Symmetrical, Dovetail, Qb/Qc = 0.5, Diverging (Idelchik 1986, Diagram 7-24)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Wye > Symmetrical > **SR5-14**



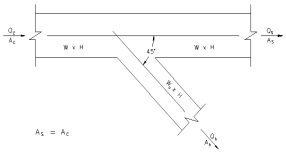
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb1**	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Wc	Width	300	12	25–2500	**Pvb1**	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Wb	Width	150	6	25–2500	**Cb1**	Loss Coefficient	(Cb1)	(Cb1)	2
Qc	Flow Rate	750	1500	5–9999999	**Pob1**	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0	**Vb2**	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
					Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
					Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
					Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
					Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Cb=fcn((Ab/Ac),1) ; Pob=Cb*Pvb`

SR5-3 Wye of the Type As + Ab > Ac, As = Ac, 45 Degree, Diverging (Idelchik 1986, Diagrams 7-15 && 7-17)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Wye > Unsymmetrical > **SR5-3**



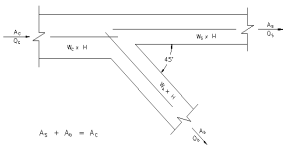
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
W	Width	300	12	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Wb	Width	150	6	25–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qc	Flow Rate	750	1500	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),2)`

SR5-8 Wye of the Type As + Ab = Ac, 45 Degree, Diverging (Idelchik 1986, Diagrams 7-16 && 7-17)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Wye > Unsymmetrical > **SR5-8**



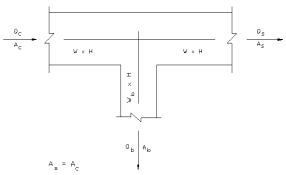
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Wc	Width	300	12	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Wb	Width	150	6	25–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Ws	Width	250	10	25–2500	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qc	Flow Rate	750	1500	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Qb	Flow Rate	150	300	5–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(As/Ac),2)`

SR5-5 Tee of the Type As + Ab > Ac, As = Ac, Diverging (Idelchik 1986, Diagrams 7-15 && 7-17)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Tee > **SR5-5**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Wb	Width	150	6	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

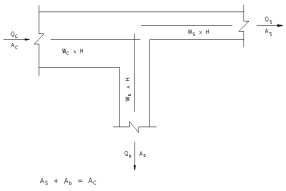
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Velocity	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),2)`

SR5-10 Tee of the Type As + Ab = Ac, Diverging (Idelchik 1986, Diagrams 7-16 && 7-17)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Tee > **SR5-10**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
Wc	Width	300	12	25–2500
Wb	Width	150	6	25–2500
Ws	Width	250	10	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

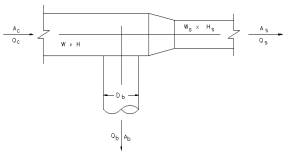
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(As/Ac),2)`

SR5-11 Tee, Rectangular Main to Round Tap, Diverging (Brooks 1993)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Tee > **SR5-11**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Hs	Height	300	10	25–2500
Ws	Width	250	10	25–2500
Db	Diameter	150	6	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	600	5–9999999
rho	Density	1.204	0.075	> 0

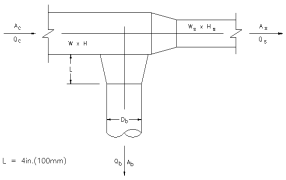
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(As/Ac),2)`

SR5-12 Tee, Rectangular Main to Round Conical Tap, Diverging (Inoue 1980; Korst 1950; Brooks 1993)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Tee > **SR5-12**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Hs	Height	300	10	25–2500
Ws	Width	250	10	25–2500
Db	Diameter	150	6	25–2500
Qc	Flow Rate	750	1500	5–9999999
Qb	Flow Rate	150	600	5–9999999
rho	Density	1.204	0.075	> 0

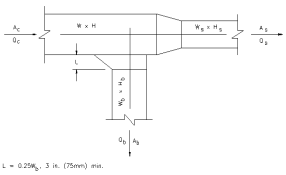
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(As/Ac),2)`

SR5-13 Tee, 45 Degree Entry Branch, Diverging (Brooks 1993)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Tee > **SR5-13**



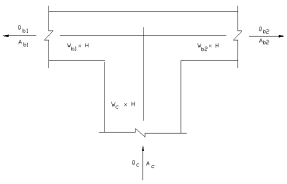
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
W	Width	300	12	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Hs	Height	300	10	25–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Ws	Width	250	10	25–2500	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Hb	Height	150	6	25–2500	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Wb	Width	150	6	25–2500	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
Qc	Flow Rate	750	1500	5–9999999	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Qb	Flow Rate	150	600	5–9999999	**Pvc**	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
rho	Density	1.204	0.075	> 0	**Cs**	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(As/Ac),2)`

SR5-15 Bullhead Tee without Vanes, Diverging (Idelchik 1986, Diagrams 7-16 && 7-30)

RECTANGULAR • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Rectangular > Junctions, Diverging > Tee > **SR5-15**



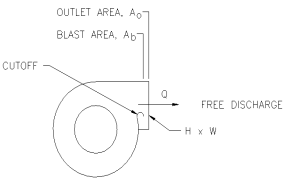
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb1**	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Wc	Width	300	12	25–2500	**Pvb1**	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Wb1	Width	200	8	25–2500	**Cb1**	Loss Coefficient	(Cb1)	(Cb1)	2
Wb2	Width	200	8	25–2500	**Pob1**	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Qc	Flow Rate	750	1500	5–9999999	**Vb2**	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Qb1	Flow Rate	150	700	5–9999999	**Pvb2**	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
rho	Density	1.204	0.075	> 0	**Cb2**	Loss Coefficient	(Cb2)	(Cb2)	2
					Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
					Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=H*Wc/ARC ; Ab1=H*Wb1/ARC ; Ab2=H*Wb2/ARC`

SR7-1 Fan, Centrifugal, without an Outlet Diffuser, Free Discharge (AMCA 1990, Fig. 8-3)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > **SR7-1**



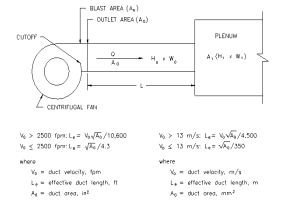
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Outlet Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Outlet Width	800	32	225–1950	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Co**	Loss Ceofficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	12000	24210	5–9999999					
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(AbOAo,1) ; Po=Co*Pv`

SR7-4 Fan, Centrifugal, Discharging into a Plenum (AMCA 1990, Fig. 8-3)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > **SR7-4**



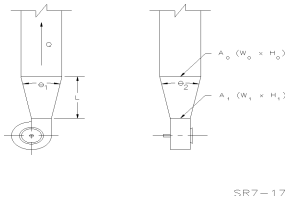
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Wo	Width	800	32	225–1950	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
H1	Height	2000	80	325–4000	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
W1	Width	1500	60	225–4000	**Pv1**	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
L	Length	300	12	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
					LOLe	L/Le Ratio:	0 to 10	0 to 10	2
Q	Flow Rate	12000	24210	5–9999999					
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pvo`

SR7-17 Pyramidal Diffuser at Centrifugal Fan Outlet with Ductwork (Idelchik 1986, Diagram 5-16)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Duffuser at Centrifugal Fan Outlet > **SR7-17**



Var	Parameter	SI Def	IP Def	Range
H1	Height	1075	43	325–2600
W1	Width	800	32	225–1950
Ho	Height	1500	60	325–4000
Wo	Width	1125	45	225–4000
Q	Flow Rate	12000	24210	5–9999999
rho	Density	1.204	0.075	> 0

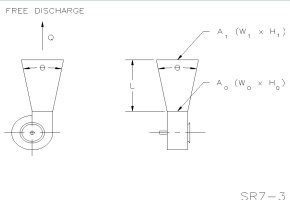
Var	Output Metric	SI Unit	IP Unit	Dec
L	Length	(L, mm)	(L, in.)	0
Theta 1	Angle	(Theta1, deg.)	(Theta1, deg.)	0
Theta 2	Angle	(Theta2, deg.)	(Theta2, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0
AoOA 1	Area Ratio (Ao/A1):	1.5 to 4.0	1.5 to 4.0	2

Governing Loss Formulation: `Co=P1/Pvo ; Po=P1`

SR7-3 Pyramidal Diffuser at Centrifugal Fan Outlet, Free Discharge (Idelchik 1986, Diagram 11-11)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Duffuser at Centrifugal Fan Outlet > **SR7-3**



Var	Parameter	SI Def	IP Def	Range
Ho	Height	1075	43	325–2600
Wo	Width	800	32	225–1950
Q	Flow Rate	12000	24210	5–9999999
rho	Density	1.204	0.075	> 0

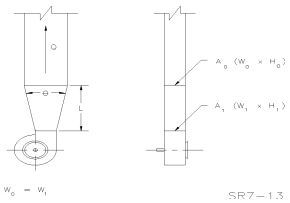
Var	Output Metric	SI Unit	IP Unit	Dec
H1	Height	(H1, mm)	(H1, in.)	0
W1	Width	(w1, mm)	(W1, in.)	0
Theta H	Theta1	(deg.)	(deg.)	0
Theta W	Theta2	(deg.)	(deg.)	0
L	Length	(L, mm)	(L, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0
A1OA o	Area Ratio	(A1/Ao)	(A1/Ao)	2

Governing Loss Formulation: `Co=fcn((A1/Ao),Theta,1) ; Po=Co*Pvo`

SR7-13 Plane Symmetric Diffuser at Centrifugal Fan Outlet with (Ductwork (Idelchik 1986, Diagram 5-12)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Duffuser at Centrifugal Fan Outlet > **SR7-13**



Var	Parameter	SI Def	IP Def	Range
H1	Height	1075	43	325–2600
W1	Width	800	32	225–1950
Ho	Height	1625	65	325–3000
Q	Flow Rate	12000	24210	5–9999999
rho	Density	1.204	0.075	> 0

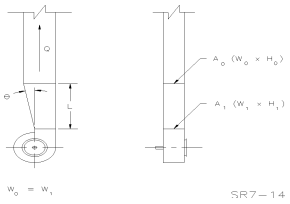
Var	Output Metric	SI Unit	IP Unit	Dec
L	Length	(L, mm)	(L, in.)	0
Theta	Theta	(deg.)	(deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0
AoOA 1	Area Ratio (Ao/A1):	1.5 to 4.0	1.5 to 4.0	2

Governing Loss Formulation: `Co=P1/Pvo ; Po=P1`

SR7-14 Plane Asymmetric Diffuser (Configuration 1) at Centrifugal (Fan Outlet with Ductwork (Idelchik 1986, Diagram 5-13)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Duffuser at Centrifugal Fan Outlet > Plane Asymmetric > **SR7-14**



Var	Parameter	SI Def	IP Def	Range
H1	Height	1075	43	325–2600
W1	Width	800	32	225–1950
Ho	Height	1625	65	325–3000
Q	Flow Rate	12000	24210	5–9999999
rho	Density	1.204	0.075	> 0

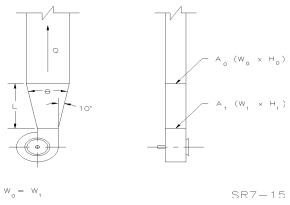
Var	Output Metric	SI Unit	IP Unit	Dec
L	Length	(L, mm)	(L, in.)	0
Theta	Theta	(deg.)	(deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0
AoOA 1	Area Ratio (Ao/A1):	1.5 to 4.0	1.5 to 4.0	2

Governing Loss Formulation: `Co=P1/Pvo ; Po=P1`

SR7-15 Plane Asymmetric Diffuser (Configuration 2) at Centrifugal (Fan Outlet with Ductwork (Idelchik 1986, Diagram 5-14))

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Duffuser at Centrifugal Fan Outlet > Plane Asymmetric > **SR7-15**



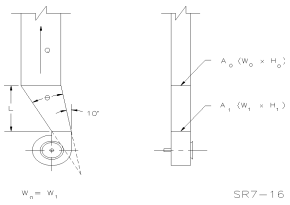
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H1	Height	1075	43	325–2600	**L**	Length	(L, mm)	(L, in.)	0
W1	Width	800	32	225–1950	**Theta**	Angle	(Theta, deg.)	(Theta, deg.)	0
Ho	Height	1625	65	325–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	12000	24210	5–9999999	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0
					AoOA1	Area Ratio (Ao/A1):	1.5 to 4.0	1.5 to 4.0	2

Governing Loss Formulation: `Co=P1/Pvo ; Po=P1`

SR7-16 Plane Asymmetric Diffuser (Configuration 3) at Centrifugal (Fan Outlet with Ductwork (Idelchik 1986, Diagram 5-15))

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Duffuser at Centrifugal Fan Outlet > Plane Asymmetric > **SR7-16**



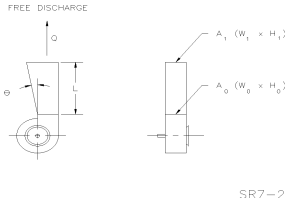
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H1	Height	1075	43	325–2600	**L**	Length	(L, mm)	(L, in.)	0
W1	Width	800	32	225–1950	**Theta**	Angle	(Theta, deg.)	(Theta, deg.)	0
Ho	Height	1625	65	325–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	12000	24210	5–9999999	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0
					AoOA1	Area Ratio (Ao/A1):	1.5 to 4.0	1.5 to 4.0	2

Governing Loss Formulation: `Co=P1/Pvo ; Po=P1`

SR7-2 Plane Asymmetric Diffuser at Centrifugal Fan Outlet (Free Discharge (Idelchik 1986, Diagram 11-11))

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Duffuser at Centrifugal Fan Outlet > Plane Asymmetric > **SR7-2**



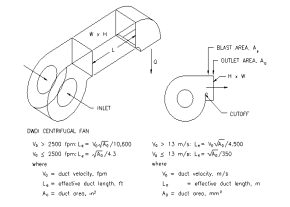
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	1075	43	325–2600	**H1**	Height	(H1, mm)	(H1, in.)	0
Wo	Width	800	32	225–1950	**W1**	Width	(W1, mm)	(W1, in.)	0
Q	Flow Rate	12000	24210	5–9999999	**Theta**	Theta	(deg.)	(deg.)	0
rho	Density	1.204	0.075	> 0	**L**	Length	(L, mm)	(L, in.)	0
					Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
					V1	Velocity	(V1, m/s)	(V1, fpm)	1
					Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0
					A1OAo	Area Ratio	(A1/Ao)	(A1/Ao)	2

Governing Loss Formulation: `Co=fcn((A1/Ao),Theta,1) ; Po=Co*Pvo`

SR7-9 Fan Outlet, Centrifugal, DWDI, with Elbow (Position A) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-9**



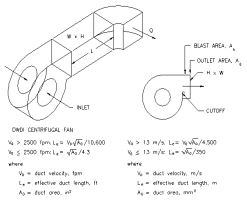
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	800	32	225–1950	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	600	24	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	12000	24210	5–9999999	**LOLe**	L/Le Ratio:	0 to 10	0 to 10	2
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pv`

SR7-10 Fan Outlet, Centrifugal, DWDI, with Elbow (Position B) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-10**



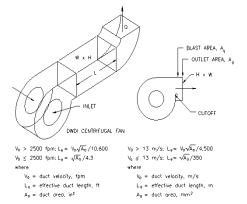
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	800	32	225–1950	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	600	24	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	12000	24210	5–9999999	**LOLe**	L/Le Ratio:	0 to 10	0 to 10	2
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((LOLe),AbOAo,1) ; Po=Co*Pv`

SR7-11 Fan Outlet, Centrifugal, DWDI, with Elbow (Position C) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-11**



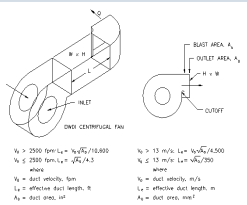
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	800	32	225–1950	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	600	24	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	12000	24210	5–9999999	**LOLe**	L/Le Ratio:	0 to 10	0 to 10	2
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pv`

SR7-12 Fan Outlet, Centrifugal, DWDI, with Elbow (Position D) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-12**



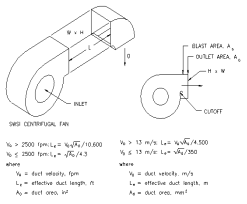
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	800	32	225–1950	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	600	24	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	12000	24210	5–9999999	**LOLe**	L/Le Ratio:	0 to 10	0 to 10	2
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pv`

SR7-5 Fan Outlet, Centrifugal, SWSI, with Elbow (Position A) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-5**



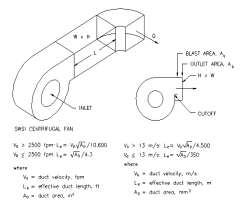
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	800	32	225–1950	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	300	12	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	12000	24210	5–9999999	**LOLe**	L/Le Ratio:	0 to 10	0 to 10	2
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pv`

SR7-6 Fan Outlet, Centrifugal, SWSI, with Elbow (Position B) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-6**



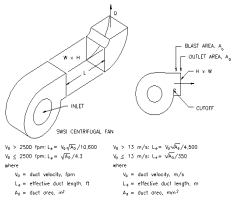
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1075	43	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	800	32	225–1950	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	600	24	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	12000	24210	5–9999999	**LOLe**	L/Le Ratio:	0 to 10	0 to 10	2
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pv`

SR7-7 Fan Outlet, Centrifugal, SWSI, with Elbow (Position C) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-7**



Var	Parameter	SI Def	IP Def	Range
H	Height	1075	43	325–2600
W	Width	800	32	225–1950
L	Length	600	24	0–6000
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1
Q	Flow Rate	12000	24210	5–9999999
rho	Density	1.204	0.075	> 0

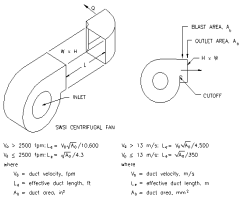
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0
LOLe	L/Le Ratio:	0 to 10	0 to 10	2

Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pv`

SR7-8 Fan Outlet, Centrifugal, SWSI, with Elbow (Position D) (AMCA 1990, Fig. 8-5)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Supply > Rectangular > Fan & System Interactions > Fan Outlet, Centrifugal > **SR7-8**



Var	Parameter	SI Def	IP Def	Range
H	Height	1075	43	325–2600
W	Width	800	32	225–1950
L	Length	600	24	0–6000
AbOAo	Blast Area Ratio	0.8	0.8	0.4–1
Q	Flow Rate	12000	24210	5–9999999
rho	Density	1.204	0.075	> 0

Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0
LOLe	L/Le Ratio:	0 to 10	0 to 10	2

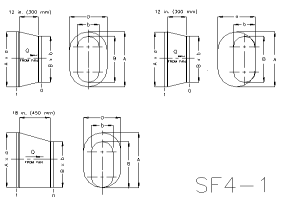
Governing Loss Formulation: `Co=fcn(LOLe,AbOAo,1) ; Po=Co*Pv`

Supply • Flat Oval Duct Catalog (5 Fittings)

SF4-1 Transition, Flat Oval, Concentric && Eccentric, Supply Air Systems (Zhang, Sun, Ford && Denny 2014)

FLAT OVAL • TRANSITIONS

Hierarchy Path: Supply > Flat Oval > Transitions > **SF4-1**



Var	Parameter	SI Def	IP Def	Range
A	Upstream Major Axi	600	24	70–2500
a	Upstream Miinor Ax	400	16	70–2500
B	Downstream Major A	300	12	70–2500
b	Downstream Minor A	200	8	70–2500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

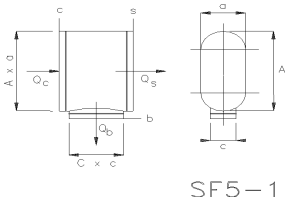
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.032*LOG(A/ADC*a/ADC/(B/ADC*b/ADC))*((A/ADC)/(a/ADC))^1.58 ; Po=Co*Pvo`

SF5-1 Tee, Flat Oval, Straight Body Branch, Diverging (Gibbs && Idem 2012)

FLAT OVAL • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Flat Oval > Junctions, Diverging > **SF5-1**



Var	Parameter	SI Def	IP Def	Range
W	Main, Major Axis	455	18	70–2500
H	Main, Minor Axis	255	10	70–2500
Qc	Flow Rate	1250	2600	5–9999999
Wb	Branch, Major Axis	230	9	70–2500
Hb	Branch, Minor Axis	100	4	70–2500
Qb	Flow Rate	150	260	0–9999999
rho	Density	1.204	0.075	> 0

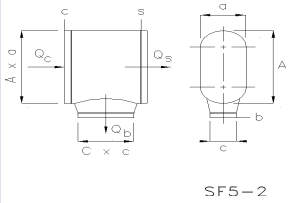
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in.wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=1.573*Qbc^-1.558*Abc^1.488 ; Cs=0.02 ; Pos=Cs* Pvs`

SF5-2 Tee, Flat Oval, Conical Body Branch, Diverging (Townsend et al. 1996; Idem && Khodabakhsh 1999)

FLAT OVAL • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Flat Oval > Junctions, Diverging > **SF5-2**



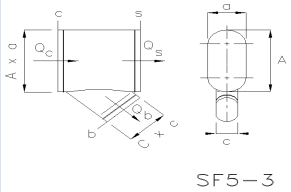
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Main, Major Axis	455	18	70–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
H	Main, Minor Axis	255	10	70–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in.wg)	0
Qc	Flow Rate	750	2600	5–9999999	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Wb	Branch, Major Axis	230	9	150–2800	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Hb	Branch, Minor Axis	100	4	70–650	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Qb	Flow Rate	150	260	0–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn(Qbc, Abc, 1) ; Pos=Cs* Pvs`

SF5-3 Wye, Flat Oval, 45 Degree, Straight Body Branch, Diverging (Gibbs && Idem 2012)

FLAT OVAL • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Flat Oval > Junctions, Diverging > **SF5-3**



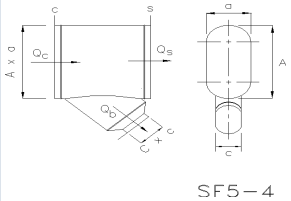
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Main, Major Axis	455	18	70–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
H	Main, Minor Axis	255	10	70–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in.wg)	0
Qc	Flow Rate	1250	2600	5–9999999	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Wb	Branch, Major Axis	230	9	70–2500	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Hb	Branch, Minor Axis	100	4	70–2500	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Qb	Flow Rate	150	260	0–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=0.702*Qbc^-1.907*Abc^1.865 ; Pos=Cs* Pvs`

SF5-4 Wye, Flat Oval, 45 Degree, Conical Body Branch, Diverging (Townsend et al. 1996; Idem && Khodabakhsh 1999)

FLAT OVAL • JUNCTIONS, DIVERGING

Hierarchy Path: Supply > Flat Oval > Junctions, Diverging > **SF5-4**



Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Main, Major Axis	455	18	70–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
H	Main, Minor Axis	255	10	70–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in.wg)	0
Qc	Flow Rate	1250	2600	5–9999999	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Wb	Branch, Major Axis	230	9	70–2500	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Hb	Branch, Minor Axis	100	4	70–2500	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Qb	Flow Rate	150	260	0–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn(Qbc, Abc, 1) ; Pos=Cs* Pvs`

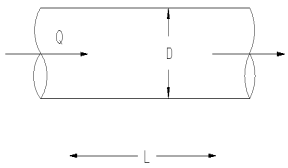
COMMON AIR SYSTEMS — EXPANDED CATALOG

Common • Round Duct Catalog (36 Fittings)

CD11-1 Straight Duct, Round (Colebrook 1939)

ROUND • FITTINGS

Hierarchy Path: Common > Round > **CD11-1**



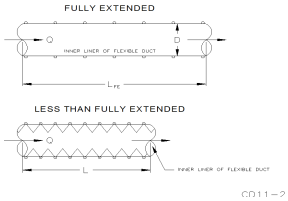
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**V**	Velocity	(V, m/s)	(V, fpm)	1
L	Length	1	100	0.1–1000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
ei	Absolute Roughness	0.09	0.0003	0.0004–4.6	**f**	Friction Factor	(f)	(f)	4
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Po=f*(L*RC/D)*Pv`

CD11-2 Straight Duct, Flexible (Abushakra 2004 and Culp 2011)

ROUND • FITTINGS

Hierarchy Path: Common > Round > **CD11-2**



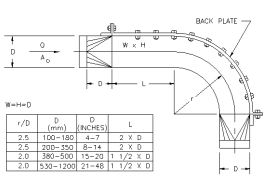
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Q	Flow Rate	500	1000	5–9999999	**V**	Velocity	(V, m/s)	(V, fpm)	1
D	Diameter	250	10	100–500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Installed Length	12.2	40	0–70	**Kc**	Compressed	(%)	(%)	0
L_FE	Length Fully Exten	15.2	50	0–70	**PDCF**	Pressure Drop Correcti	(PDCF)	(PDCF)	1
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Po=f*(L*RC/D)*Pv ; Po=PDCF*Po`

CD3-11 Elbow, Flat-back, 90 Degree (UMC 1985, Report SRF1285)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > **CD3-11**



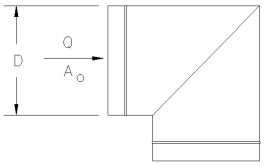
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1200	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-15 Elbow, Mitered, 90 Degree (UMC 1985, Report SRF1285)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Mitered > **CD3-15**



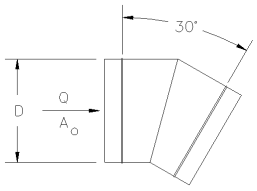
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-18 Elbow, Mitered, 30 Degree ([UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1 (Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Mitered > **CD3-18**



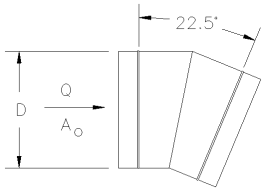
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-19 Elbow, Mitered, 22.5 Degree ([UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Mitered > **CD3-19**



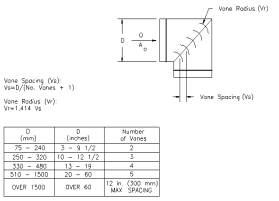
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-20 Elbow, Mitered with Vanes, 90 Degree (UMC 1985, Report SRF1285)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Mitered > **CD3-20**



D (mm)	D (inches)	Number of Vanes
75 – 240	3 – 9 1/2	2
250 – 320	10 – 12 1/2	3
330 – 480	13 – 19	4
510 – 1500	20 – 60	5
OVER 1500	OVER 60	12 in (300 mm) MAX SPACING

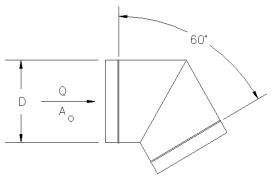
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-16 Elbow, Mitered, 60 Degree ([UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1), Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Mitered > **CD3-16**



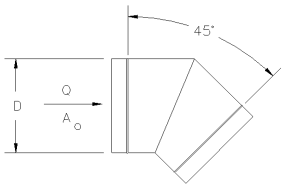
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-17 Elbow, Mitered, 45 Degree ([UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Mitered > **CD3-17**



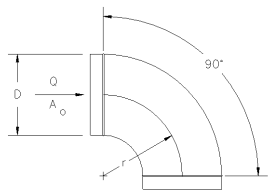
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-1 Elbow, Die Stamped, 90 Degree, r/D = 1.5 (UMC 1985, Report SRF785)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Die Stamped > **CD3-1**



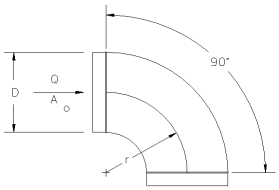
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	200	8	75–250	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-2 Elbow, Die Stamped, 90 Degree, r/D = 1.0 (UMC 1985, Report SRF785)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Die Stamped > **CD3-2**



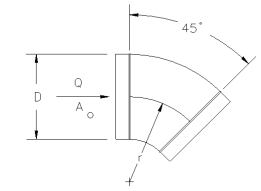
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	200	8	75–250	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-3 Elbow, Die Stamped, 45 Degree, r/D = 1.5 ([UMC 1985, Report SRF785; Idelchik 1986, Diagram 6-1, Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Die Stamped > **CD3-3**



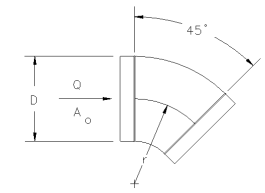
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	200	8	75–250	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-4 Elbow, Die Stamped, 45 Degree, r/D = 1.0 ([UMC 1985, Report SRF785; Idelchik 1986, Diagram 6-1, Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Die Stamped > **CD3-4**



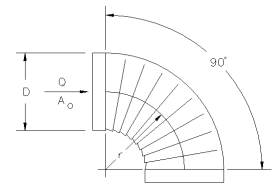
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	200	8	75–250	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-5 Elbow, Pleated, 90 Degree, r/D = 1.5 (UMC 1989, Report SRS189)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Pleated > **CD3-5**



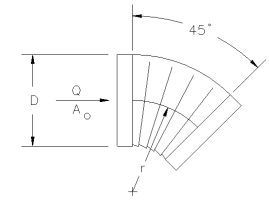
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	100–400	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-7 Elbow, Pleated, 45 Degree, r/D = 1.5 (UMC 1989, Report SRS189)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Pleated > **CD3-7**



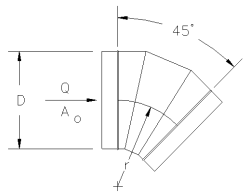
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	100–400	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-14 Elbow, 3 Gore, 45 Degree, r/D = 1.5 ([UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Gored > **CD3-14**



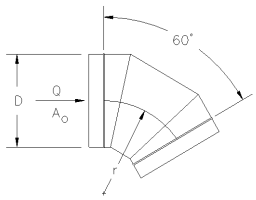
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-13 Elbow, 3 Gore, 60 Degree, r/D = 1.5 ([UMC 1985, Report SRF1285; Idelchik 1986, Diagram 6-1, Graph (a)])

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Gored > **CD3-13**



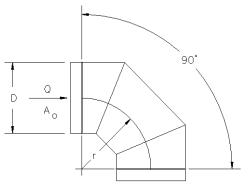
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-12 Elbow, 3 Gore, 90 Degree, r/D=0.75 to 2.0 (Locklin 1950, Fig. 10)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Gored > **CD3-12**



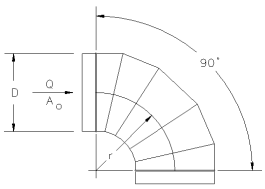
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
RDR	Radius/Diameter Ra	1	1	0.75–2	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(RDR,1) ; Po=Co*Pv`

CD3-9 Elbow, 5 Gore, 90 Degree, r/D = 1.5 (UMC 1985, Report SRF1285)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Gored > **CD3-9**



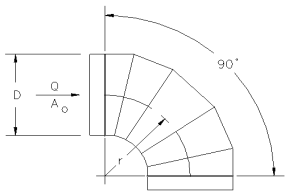
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-21 Elbow, 5 Gore, 90 Degree, r/D = 1.0 (UMC 1985, Report SRF1285; Brooks 1993)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Gored > **CD3-21**



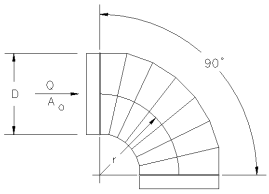
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD3-10 Elbow, 7 Gore, 90 Degree, r/D = 2.5 (UMC 1985, Report SRF1285)

ROUND • ELBOWS

Hierarchy Path: Common > Round > Elbows > Gored > **CD3-10**



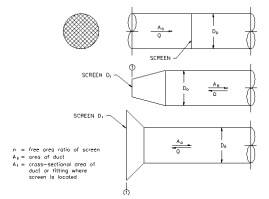
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–1500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(D/DC,1) ; Po=Co*Pv`

CD6-1 Screen (Only) (Idelchik 1986, Diagrams 8-6 and 3-15)

ROUND • OBSTRUCTIONS

Hierarchy Path: Common > Round > Obstructions > **CD6-1**



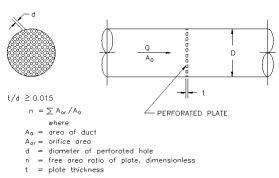
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	75–3000	**OA**	Open Area	(%)	(%)	0
D1	Diameter	450	18	75–3000	**AR**	Area Ratio	(A1/Ao)	(A1/Ao)	2
n	Free Area Ratio	0.8	0.8	0.05–1	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(n,AR,1) ; Po=Co*Pv`

CD6-2 Perforated Plate, Thick (Idelchik 1986, Diagram 8-3)

ROUND • OBSTRUCTIONS

Hierarchy Path: Common > Round > Obstructions > **CD6-2**



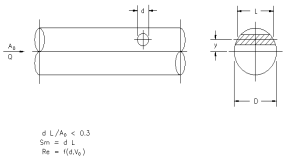
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**OA**	Open Area	(%)	(%)	0
n	Free Area Ratio	0.8	0.8	0.02–1	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
d	Hole Diameter	25	1	6–100	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
t	Plate Thickness	6	0.25	4–14	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0	**td**	t/d	-	-	3

Governing Loss Formulation: `Co=fcn(n,(t/d),1) ; Po=Co*Pv`

CD6-3 Obstruction, Smooth Cylinder in Round Duct (Idelchik 1986, Diagram 10-1)

ROUND • OBSTRUCTIONS

Hierarchy Path: Common > Round > Obstructions > **CD6-3**



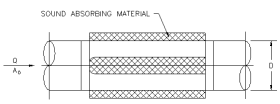
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
dc	Diameter	25	1	3–250	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
y	Distance	0	0	0–1950	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0	**z**	Sm/Ao	-	-	2

Governing Loss Formulation: `Co=fcn((Sm/A),Re,(y/D),1) ; Po=Co*Pv`

CD8-1 Silencer, Ducted, Double Wall with Bullet (Standard (High) Pressure Drop (see Table below)

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Double Wall > **CD8-1**



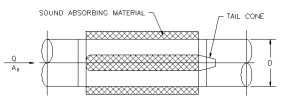
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	7500	15000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.81 ; Po=Co*Pv`

CD8-2 Silencer, Ducted, Double Wall with Bullet and Tail Cone (Standard (High) Pressure Drop (see Table below))

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Double Wall > **CD8-2**



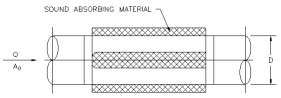
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	7500	15000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.67 ; Po=Co*Pv`

CD8-3 Silencer, Ducted, Double Wall with Bullet (Low Pressure Drop (see Table below))

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Double Wall > **CD8-3**



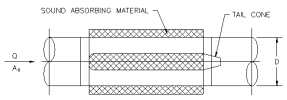
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	9000	18000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.29 ; Po=Co*Pv`

CD8-4 Silencer, Ducted, Double Wall with Bullet and Tail Cone (Low Pressure Drop (see Table below))

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Double Wall > **CD8-4**



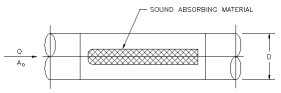
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	9000	18000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.23 ; Po=Co*Pv`

CD8-5 Silencer, Ducted, Single Wall with Bullet (Standard (High) Pressure Drop)

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Single Wall > **CD8-5**



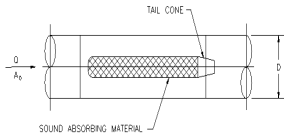
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	7500	15000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.87 ; Po=Co*Pv`

CD8-6 Silencer, Ducted, Single Wall with Bullet and Tail Cone (Standard (High) Pressure Drop)

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Single Wall > **CD8-6**



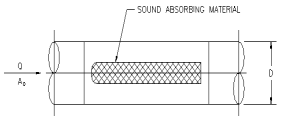
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	7500	15000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.78 ; Po=Co*Pv`

CD8-7 Silencer, Ducted, Single Wall with Bullet (Low Pressure Drop (see Table below))

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Single Wall > **CD8-7**



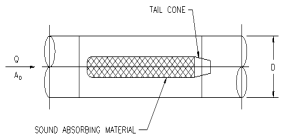
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	9000	18000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.44 ; Po=Co*Pv`

CD8-8 Silencer, Ducted, Single Wall with Bullet and Tail Cone (Low Pressure Drop (see Table below))

ROUND • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Round > Duct-Mounted Equipment > Silencer, Single Wall > **CD8-8**



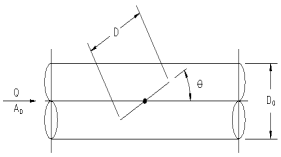
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	900	36	300–1500	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	9000	18000	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.44 ; Po=Co*Pv`

CD9-1 Damper, Butterfly (Idelchik 1986, Diagram 9-16)

ROUND • DAMPERS

Hierarchy Path: Common > Round > Dampers > **CD9-1**



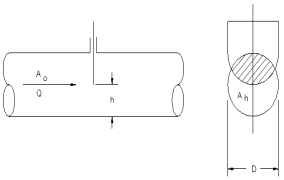
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	100–500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Theta	Angle	0	0	0–90	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,1) ; Po=Co*Pv`

CD9-2 Damper, Gate (Idelchik 1986, Diagram 9-5)

ROUND • DAMPERS

Hierarchy Path: Common > Round > Dampers > **CD9-2**



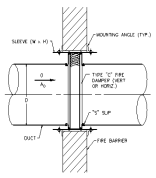
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
hg	Height	150	6	0–500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((hg/D),1) ; Po=Co*Pv`

CD9-3 Fire Damper, Curtain Type, Style C, Horizontal Duct (Vertical Installation) (1-1/2 && 3 hour rated; For use in dynamic && static systems)

ROUND • DAMPERS

Hierarchy Path: Common > Round > Dampers > **CD9-3**



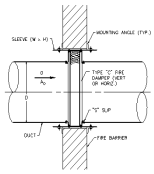
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	300	12	76–1118	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.12 ; Po=Co*Pv`

CD9-4 Fire Damper, Curtain Type, Style C, Vertical Duct (Horizontal Installation) (1-1/2 && 3 hour rated; For use in dynamic && static systems)

ROUND • DAMPERS

Hierarchy Path: Common > Round > Dampers > **CD9-4**



Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Duct Diameter	300	12	76–2159	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

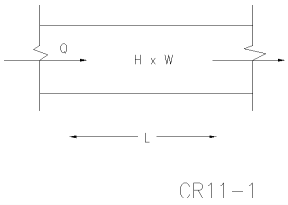
Governing Loss Formulation: `Co=0.12 ; Po=Co*Pv`

Common • Rectangular Duct Catalog (44 Fittings)

CR11-1 Straight Duct, Rectangular (Huebscher 1948)

RECTANGULAR • FITTINGS

Hierarchy Path: Common > Rectangular > **CR11-1**



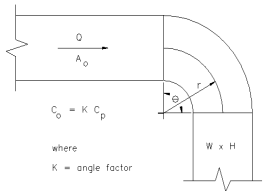
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	75–4000	**V**	Velocity	(V, m/s)	(V, fpm)	1
W	Width	300	12	75–4000	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in. wg)	0
ei	Absolute Roughness	0.09	0.0003	0.01–4.6	**f**	Friction Factor	(f)	(f)	4
L	Length	1	100	0.1–1000	**Po**	**Pressure Loss**	(Pa)	(in. wg)	1
Q	Flow Rate	700	1500	5–9999999					
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Po=f*(L*RC/Dh)*Pv`

CR3-1 Elbow, Smooth Radius without Vanes (Idelchik 1986, Diagram 6-1)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Smooth Radius > **CR3-1**



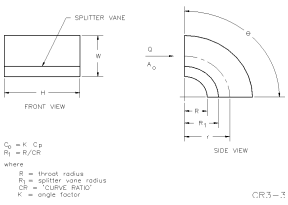
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
H	Height	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
RR	Radius Ratio	1.5	1.5	0.5–2	**Co**	Loss Coefficient	(Co)	(Co)	2
Theta	Angle	90	90	0–180	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	700	1500	5–9999999					
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Cp=fcn((H/W),RR,1) ; K=fcn(Theta,2) ; Co=Cp*K`

CR3-3 Elbow, Smooth Radius, One Splitter Vane (Locklin 1950; Madison && Parker 1936)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Smooth Radius > **CR3-3**



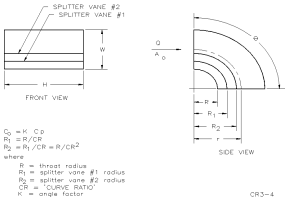
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**R**	Throat Radius	(R, mm)	(R, in.)	0
H	Height	300	12	75–4000	**R1**	Splitter Vane Radius	(R1, mm)	(R1, in.)	0
RR	Radius Ratio	1	1	0.55–1	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Theta	Angle	90	90	0–90	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cp=fcn((H/W),(RR),1) ; K=fcn(Theta,2) ; Co=Cp*K`

CR3-4 Elbow, Smooth Radius, Two Splitter Vanes (Locklin 1950; Madison && Parker 1936)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Smooth Radius > **CR3-4**



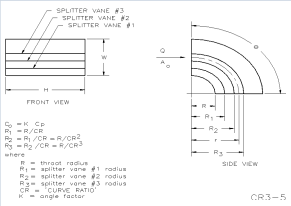
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**R**	Throat Radius	(R, mm)	(R, in.)	0
H	Height	300	12	75–4000	**R1**	Splitter Vane Radius	(R1, mm)	(R1, in.)	0
RR	Radius Ratio	1	1	0.55–1	**R2**	Splitter Vane Radius	(R2, mm)	(R2, in.)	0
Theta	Angle	90	90	0–90	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cp=fcn((H/W),(RR),1) ; K=fcn(Theta,2) ; Co=Cp*K`

CR3-5 Elbow, Smooth Radius, Three Splitter Vanes (Locklin 1950; Madison && Parker 1936)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Smooth Radius > **CR3-5**



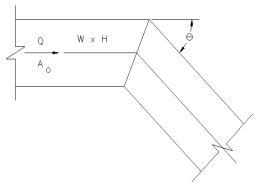
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**R**	Throat Radius	(R, mm)	(R, in.)	0
H	Height	300	12	75–4000	**R1**	Splitter Vane Radius	(R1, mm)	(R1, in.)	0
RR	Radius Radius	1	1	0.55–1	**R2**	Splitter Vane Radius	(R2, mm)	(R2, in.)	0
Theta	Angle	90	90	0–90	**R3**	Splitter Vane Radius	(R3, mm)	(R3, in.)	0
Q	Flow Rate	700	1500	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cp=fcn((H/W),(RR),1) ; K=fcn(Theta,2) ; Co=Cp*K`

CR3-6 Elbow, Mitered (Idelchik 1986, Diagram 6-5)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Mitered > **CR3-6**



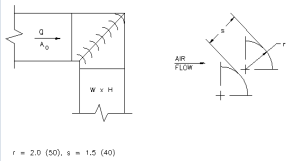
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
H	Height	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Theta	Angle	90	90	20–90	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((H/W),Theta,1) ; Po=Co*Pv`

CR3-9 Elbow, Mitered, 90 Degree, Single-Thickness Vanes (1 1/2-in. Vane Spacing (Brooks 1993)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Mitered > Single-Thickness Vanes > **CR3-9**



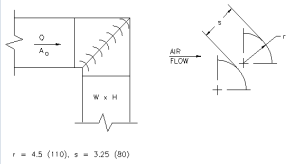
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
H	Height	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.11 ; Po=Co*Pv`

CR3-12 Elbow, Mitered, 90 Degree, Single-Thickness Vanes (3 1/4-in. Vane Spacing (Brooks 1993)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Mitered > Single-Thickness Vanes > **CR3-12**



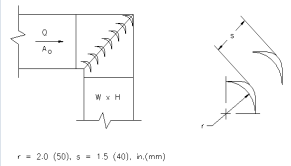
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**r**	Vane Radius	(r, mm)	(r, in.)	0
H	Height	300	12	75–4000	**s**	Vane Spacing	(s, mm)	(s, in.)	0
Q	Flow Rate	700	1500	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.33 ; Po=Co*Pv`

CR3-14 Elbow, Mitered, 90 Degree, Double-Thickness Vanes (1 1/2-in. Vane Spacing (Brooks 1993)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Mitered > Double-Thickness Vanes > **CR3-14**



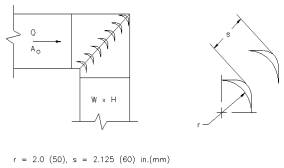
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**r**	Vane Radius	(r, mm)	(r, in.)	0
H	Height	300	12	75–4000	**s**	Vane Spacing	(s, mm)	(s, in.)	0
Q	Flow Rate	700	1500	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.38 ; Po=Co*Pv`

CR3-15 Elbow, Mitered, 90 Degree, Double-Thickness Vanes (2 1/8-in. Vane Spacing (Rozell 1974))

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Mitered > Double-Thickness Vanes > **CR3-15**



Var	Parameter	SI Def	IP Def	Range
W	Width	300	12	75–4000
H	Height	300	12	75–4000
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

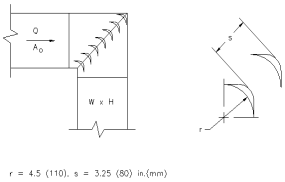
Var	Output Metric	SI Unit	IP Unit	Dec
r	Vane Radius	(r, mm)	(r, in.)	0
s	Vane Spacing	(s, mm)	(s, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.25 ; Po=Co*Pv`

CR3-16 Elbow, Mitered, 90 Degree, Double-Thickness Vanes (3 1/4-in. Vane Spacing (Brooks 1993))

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Mitered > Double-Thickness Vanes > **CR3-16**



Var	Parameter	SI Def	IP Def	Range
W	Width	300	12	75–4000
H	Height	300	12	75–4000
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

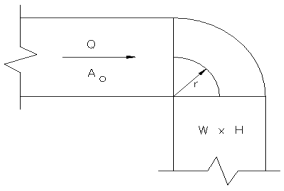
Var	Output Metric	SI Unit	IP Unit	Dec
r	Vane Radius	(r, mm)	(r, in.)	0
s	Vane Spacing	(s, mm)	(s, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.41 ; Po=Co*Pv`

CR3-2 Elbow, Sharp Throat (r/W=0.5), Radius Heel, 90 Degree (Idelchik 1986, Diagram 6-1)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Other Elbows > **CR3-2**



Var	Parameter	SI Def	IP Def	Range
W	Width	300	12	75–4000
H	Height	300	12	75–4000
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

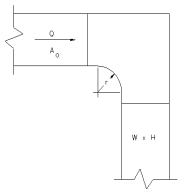
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((H/W),1) ; Po=Co*Pv`

CR3-7 Elbow, Rounded Throat, Sharp Heel, 90 Degree (Idelchik 1986, Diagram 6-8)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Other Elbows > **CR3-7**



Var	Parameter	SI Def	IP Def	Range
W	Width	300	12	75–4000
H	Height	300	12	75–4000
r	Throat Radius	150	6	25–1500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

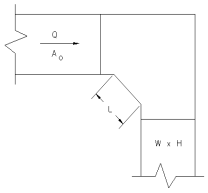
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((H/W),(r/W),1) ; Po=Co*Pv`

CR3-8 Elbow, Beveled Throat, Sharp Heel, 90 Degree (Idelchik 1986, Diagram 6-8)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Other Elbows > **CR3-8**



Var	Parameter	SI Def	IP Def	Range
W	Width	300	12	75–4000
H	Height	300	12	75–4000
L	Length	150	6	25–1500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

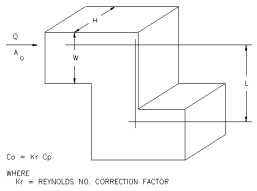
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((H/W),(L/W),1) ; Po=Co*Pv`

CR3-17 Elbow, Z-Shape (Idelchik 1986, Diagram 6-11)

RECTANGULAR • ELBOWS

Hierarchy Path: Common > Rectangular > Elbows > Other Elbows > **CR3-17**



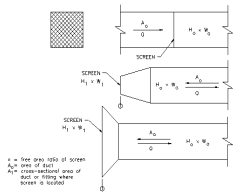
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Width	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
H	Height	600	24	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Offset	300	12	0–3000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	1400	3000	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Cp=fcn(L/W,H/W,1) ; Co=Cp*Kr ; Po=Co*Pv`

CR6-1 Screen (Only) (Idelchik 1986, Diagrams 8-6 and 3-15)

RECTANGULAR • OBSTRUCTIONS

Hierarchy Path: Common > Rectangular > Obstructions > **CR6-1**



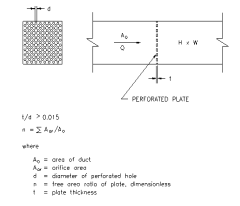
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	300	12	75–4000	**OA**	Open Area	(%)	(%)	0
Wo	Width	300	12	75–4000	**AR**	Area Ratio	(A1/Ao)	(A1/Ao)	2
H1	Height	450	18	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W1	Width	450	18	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
n	Free Area Ratio	0.5	0.5	0.05–1	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	1000	2000	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(n,AR,1) ; Po=Co*Pv`

CR6-2 Perforated Plate, Thick (Idelchik 1986, Diagram 8-3)

RECTANGULAR • OBSTRUCTIONS

Hierarchy Path: Common > Rectangular > Obstructions > **CR6-2**



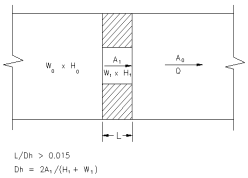
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	75–4000	**OA**	Open Area	(%)	(%)	0
W	Width	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
n	Free Area Ratio	0.8	0.8	0.02–1	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
d	Hole Diameter	25	1	6–100	**Co**	Loss Coefficient	(Co)	(Co)	2
t	Plate Thickness	6	0.25	4–14	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**td**	t/d	-	-	3
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(n,(t/d),1) ; Po=Co*Pv`

CR6-3 Orifice, Thick-Edged (Idelchik 1986, Diagram 4-15)

RECTANGULAR • OBSTRUCTIONS

Hierarchy Path: Common > Rectangular > Obstructions > **CR6-3**



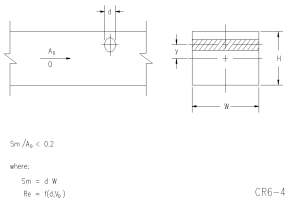
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ho	Height	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Wo	Width	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
H1	Height	150	6	75–4000	**Co**	Loss Coefficient	(Co)	(Co)	2
W1	Width	150	6	75–4000	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
L	Length	12	0.5	3–50	**LDh**	L/Dh	-	-	3
Q	Flow Rate	700	1500	5–9999999					
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((A1/Ao),(L/Dh),1) ; Po=Co*Pv`

CR6-4 Obstruction, Smooth Cylinder in Rectangular Duct (Idelchik 1986, Diagram 10-1)

RECTANGULAR • OBSTRUCTIONS

Hierarchy Path: Common > Rectangular > Obstructions > **CR6-4**



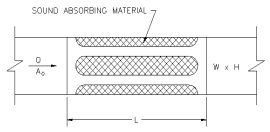
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
dc	Diameter	25	1	3–250	**Co**	Loss Coefficient	(Co)	(Co)	2
y	Distance	0	0	0–1950	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**z**	Sm/Ao	-	-	2
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((Sm/A),Re,(y/H),1) ; Po=Co*Pv`

CR8-1 Silencer, Ducted, Standard Pressure Drop (see Table below)

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Silencer > **CR8-1**



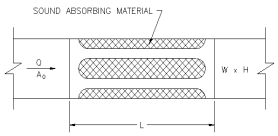
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	600	24	300–4000	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Diameter	300	12	150–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	1500	60	900–3000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	750	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(L/DC,1) ; Po=Co*Pv`

CR8-2 Silencer, Ducted, Medium Pressure Drop (see Table below)

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Silencer > **CR8-2**



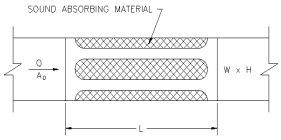
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	600	24	300–4000	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	375	15	185–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	1500	60	900–3000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	1500	3000	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(L/DC,1) ; Po=Co*Pv`

CR8-3 Silencer, Ducted, Medium-Low Pressure Drop (see Table below)

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Silencer > **CR8-3**



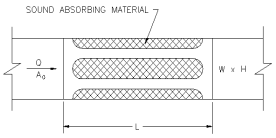
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	600	24	350–4000	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	300	12	150–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	1500	60	900–3000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	1750	3500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(L/DC,1) ; Po=Co*Pv`

CR8-4 Silencer, Ducted, Low Pressure Drop (see Table below)

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Silencer > **CR8-4**



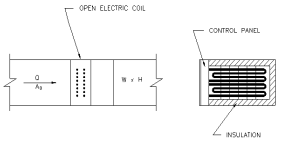
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	600	24	300–4000	**Vo**	Face Area Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	450	18	225–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	1500	60	900–3000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	2500	5000	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(L/DC,1) ; Po=Co*Pv`

CR8-5 Coil, Electric, Open Coil

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Coil > **CR8-5**



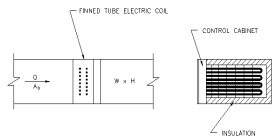
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	300	12	25–2500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	300	12	25–2500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=1.38 ; Po=Co*Pv`

CR8-6 Coil, Electric, Finned Tubular

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Coil > **CR8-6**



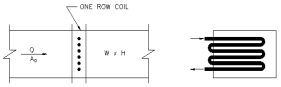
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	300	12	25–2500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	300	12	25–2500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=6.14 ; Po=Co*Pv`

CR8-7 Coil, Hot Water, One Row

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Coil > **CR8-7**



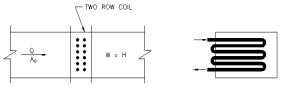
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	300	12	25–2500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	300	12	25–2500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=4.6 ; Po=Co*Pv`

CR8-8 Coil, Hot Water, Two Row

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Coil > **CR8-8**



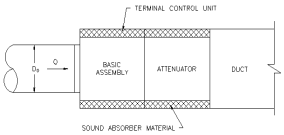
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	300	12	25–2500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	300	12	25–2500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=9.5 ; Po=Co*Pv`

CR8-9 Terminal Control Unit, Basic Assembly and Attenuator

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Terminal Unit (Box) > **CR8-9**



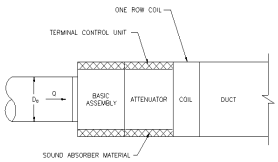
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	100–600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=2.08 ; Po=Co*Pv`

CR8-10 Terminal Control Unit, Basic Assembly, One-Row Coil and Attenuator

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Terminal Unit (Box) > **CR8-10**



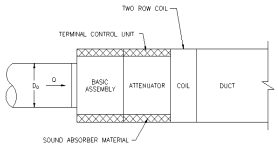
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	100–600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=2.92 ; Po=Co*Pv`

CR8-11 Terminal Control Unit, Basic Assembly, Two-Row Coil and Attenuator

RECTANGULAR • DUCT-MOUNTED EQUIPMENT

Hierarchy Path: Common > Rectangular > Duct-Mounted Equipment > Terminal Unit (Box) > **CR8-11**



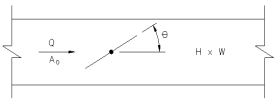
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	100–600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=3.80 ; Po=Co*Pv`

CR9-1 Damper, Butterfly (Idelchik 1986, Diagram 9-17)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-1**



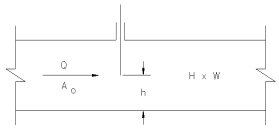
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	100–300	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	600	24	125–600	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Theta	Angle	0	0	0–90	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn(Theta,H/W,1) ; Po=Co*Pv`

CR9-2 Damper, Gate (Idelchik 1986, Diagram 9-5)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-2**



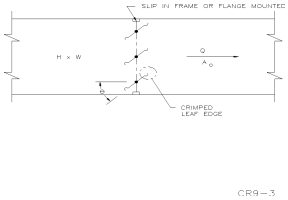
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	75–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	300	12	75–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
hg	Height	150	6	0–4000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((hg/H),(H/W),1) ; Po=Co*Pv`

CR9-3 Damper, Parallel && Opposed 3V Blades, Open (Van Becelaere && Sauer 2004)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-3**



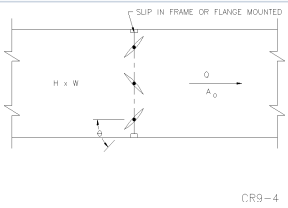
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	152–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	300	12	152–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.37 ; Po=Co*Pv`

CR9-4 Damper, Parallel && Opposed Airfoil Blades, Open (Van Becelaere && Sauer 2004)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-4**



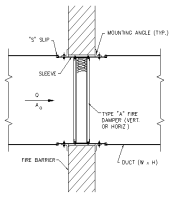
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	152–4000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	300	12	152–4000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.18 ; Po=Co*Pv`

CR9-5 Fire Damper, Curtain Type, Style A, Horizontal Duct (Vertical Installation) (1-1/2 && 3 Hour; For Use in Dynamic && Static Systems)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-5**



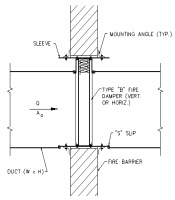
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	914	36	102–1829	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	838	33	102–2286	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.46 ; Po=Co*Pv`

CR9-6 Fire Damper, Curtain Type, Style B, Horizontal Duct (Vertical Installation) (1-1/2 && 3 Hour; For Use in Dynamic && Static Systems)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-6**



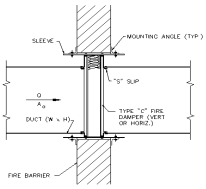
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	813	32	102–1753	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	838	33	102–2286	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.19 ; Po=Co*Pv`

CR9-7 Fire Damper, Curtain Type, Style C, Horizontal Duct (Vertical Installation) (1-1/2 && 3 Hour; For Use in Dynamic && Static Systems)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-7**



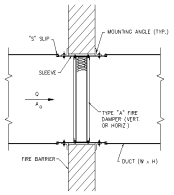
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	787	31	76–1727	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	787	31	102–2235	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.12 ; Po=Co*Pv`

CR9-8 Fire Damper, Curtain Type, Style A, Vertical Duct (Horizontal Installation) (1-1/2 && 3 Hour; For Use in Dynamic && Static Systems)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-8**



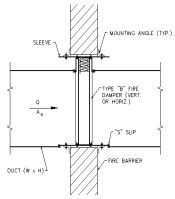
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	1219	48	152–2311	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	914	36	152–2286	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.46 ; Po=Co*Pv`

CR9-9 Fire Damper, Curtain Type, Style B, Vertical Duct (Horizontal Installation) (1-1/2 && 3 Hour; For Use in Dynamic && Static Systems)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-9**



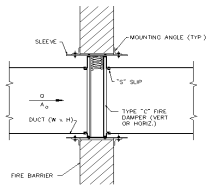
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	1067	42	102–1067	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	914	36	152–914	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.19 ; Po=Co*Pv`

CR9-10 Fire Damper, Curtain Type, Style C, Vertical Duct (Horiontal Installation) (1-1/2 && 3 Hour; For Use in Dynamoc && Static Systems)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-10**



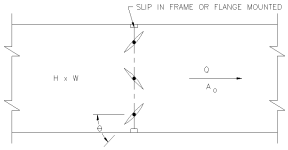
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Duct Height	1067	42	102–1067	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Duct Width	864	34	152–864	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.12 ; Po=Co*Pv`

CR9-11 Fire/Smoke Damper, 3 Hour, Airfoil Blades, Vertical Installation (Van Becelaere && Sauer 2004)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-11**



CR9–4

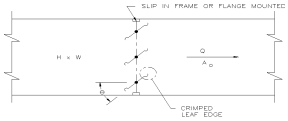
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1219	48	152–2438	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	762	30	203–3048	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.18 ; Po=Co*Pv`

CR9-12 Fire/Smoke Damper, 1-1/2 Hour, 3V Blades, Vertical Installation (Van Becelaere && Sauer 2004)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-12**



CR9–3

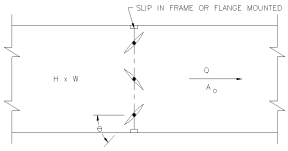
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1219	48	152–2438	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	813	32	203–3048	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.37 ; Po=Co*Pv`

CR9-13 Fire/Smoke Damper, 3 Hour, Airfoil Blades, Horizontal Installation (Van Becelaere && Sauer 2004)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-13**



CR9–4

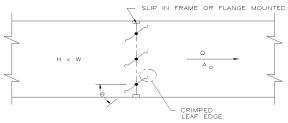
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1219	48	152–2438	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	762	30	203–3658	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.18 ; Po=Co*Pv`

CR9-14 Fire/Smoke Damper, 1-1/2 Hour, 3V Blades, Horizontal Installation (Van Becelaere && Sauer 2004)

RECTANGULAR • DAMPERS

Hierarchy Path: Common > Rectangular > Dampers > **CR9-14**



CR9–3

Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	1219	48	152–2438	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
W	Width	813	32	203–3658	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	4500	9000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

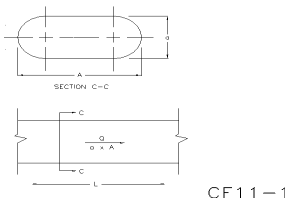
Governing Loss Formulation: `Co=0.37 ; Po=Co*Pv`

Common • Flat Oval Duct Catalog (7 Fittings)

CF11-1 Straight Duct, Flat Oval (Heyt and Diaz 1975; Townsend et al. 1994)

FLAT OVAL • FITTINGS

Hierarchy Path: Common > Flat Oval > **CF11-1**



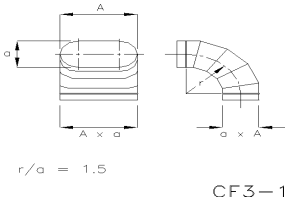
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Major Axis	960	38	70–2500	**V**	Velocity	(V, m/s)	(V, fpm)	1
H	Minor Axis	350	14	70–2500	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in. wg)	0
ei	Absolute Roughness	0.09	0.0003	0.01–4.6	**f**	Friction Factor	(f)	(f)	4
L	Length	1	100	0.1–1000	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
Q	Flow Rate	6000	12000	5–9999999					
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Po=f*(L*RC/D)*Pv`

CF3-1 Elbow, Flat Oval, 5 Gore, Easy Bend, 90 Degree (Kulkarni, Khaire and Idem 2009)

FLAT OVAL • ELBOWS

Hierarchy Path: Common > Flat Oval > Elbows > **CF3-1**



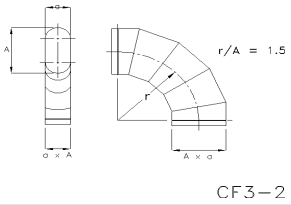
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
a2	Major Axis	960	38	70–2500	**V**	Velocity	(V, m/s)	(V, fpm)	1
a1	Minor Axis	350	14	70–2500	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in.wg)	0
Q	Flow Rate	3000	10000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
					r	Centerline Radius	(r, mm)	(r, in.)	0

Governing Loss Formulation: `Co=0.439*DhPrime^-0.065*AR^0.075 ; Po=Co*Pv`

CF3-2 Elbow, Flat Oval, 5 Gore, Hard Bend, 90 Degree (Townsend et al. 1996 and Smith && Jones 1976)

FLAT OVAL • ELBOWS

Hierarchy Path: Common > Flat Oval > Elbows > **CF3-2**



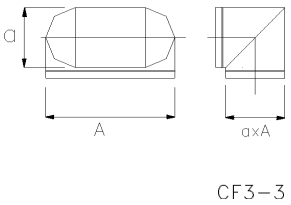
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
a2	Major Axis	960	38	70–2500	**V**	Velocity	(V, m/s)	(V, fpm)	1
a1	Minor Axis	350	14	70–2500	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in.wg)	0
Q	Flow Rate	3000	10000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
					r	Centerline Radius	(r, mm)	(r, in.)	0

Governing Loss Formulation: `Co=0.301*DhPrime^0.072*AR^0.427 ; Po=Co*Pv`

CF3-3 Elbow, Flat Oval, Mitered, Easy Bend (No Vanes), 90 Degree (Kulkarni, Khaire and Idem 2009)

FLAT OVAL • ELBOWS

Hierarchy Path: Common > Flat Oval > Elbows > **CF3-3**



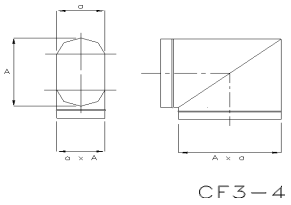
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
a2	Major Axis	960	38	70–2500	**V**	Velocity	(V, m/s)	(V, fpm)	1
a1	Minor Axis	350	14	70–2500	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in.wg)	0
Q	Flow Rate	3000	10000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=1.366*DhPrime^-0.156*AR^0.040 ; Po=Co*Pv`

CF3-4 Elbow, Flat Oval, Mitered, Hard Bend (No Vanes), 90 Degree (Kulkarni, Khaire and Idem 2009)

FLAT OVAL • ELBOWS

Hierarchy Path: Common > Flat Oval > Elbows > **CF3-4**



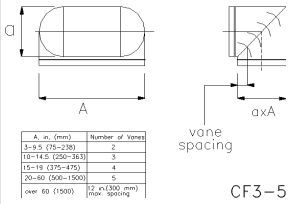
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
a2	Major Axis	960	38	70–2500	**V**	Velocity	(V, m/s)	(V, fpm)	1
a1	Minor Axis	350	14	70–2500	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in.wg)	0
Q	Flow Rate	3000	10000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=1.341*DhPrime^0.011*AR^-0.325 ; Po=Co*Pv`

CF3-5 Elbow, Flat Oval, Mitered, Easy Bend with Vanes, 90 Degree (Kulkarni, Khaire and Idem 2009)

FLAT OVAL • ELBOWS

Hierarchy Path: Common > Flat Oval > Elbows > **CF3-5**



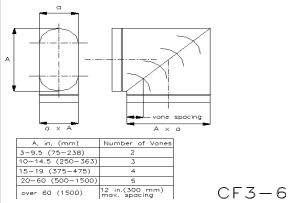
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
a2	Major Axis	960	38	70–2500	**V**	Velocity	(V, m/s)	(V, fpm)	1
a1	Minor Axis	350	14	70–2500	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in.wg)	0
Q	Flow Rate	3000	10000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.581*DhPrime^-0.647*AR^-0.0004 ; Po=Co*Pv`

CF3-6 Elbow, Flat Oval, Mitered, Hard Bend with Vanes, 90 Degree (Kulkarni, Khaire and Idem 2009)

FLAT OVAL • ELBOWS

Hierarchy Path: Common > Flat Oval > Elbows > **CF3-6**



Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
a2	Major Axis	960	38	70–2500	**V**	Velocity	(V, m/s)	(V, fpm)	1
a1	Minor Axis	350	14	70–2500	**Pv**	Vel Pres at V	(Pv, Pa)	(Pv, in.wg)	0
Q	Flow Rate	3000	10000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=0.760*DhPrime^-0.178*AR^-0.141 ; Po=Co*Pv`

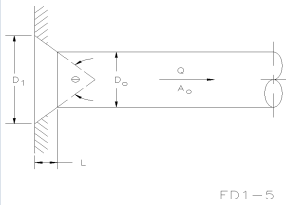
EXHAUST/RETURN AIR SYSTEMS — EXPANDED CATALOG

Exhaust/Return • Round Duct Catalog (32 Fittings)

ED1-5 Conical Bellmouth/Sudden Contraction, with Wall (Idelchik 1986, Diagram 3-7)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-5**



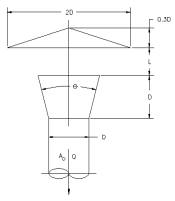
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	75–3000	**XL**	Length	(L, mm)	(L, in.)	0
Theta	Angle	60	60	0–180	**Theta 2**	Angle	(Theta, deg.)	(Theta, deg.)	0
L	Length	175	7	0–1500	**D1**	Diameter	(D1, mm)	(D1, in.)	0
Q	Flow Rate	700	1500	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta2,(XL/Do),1) ; Po=Co*Pv`

ED1-6 Intake Hood (Idelchik 1986, Diagram 3-18)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-6**



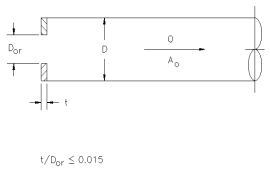
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Cap Offset	150	6	25–1500	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Theta	Angle	0	10	0–15	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((L/D),Theta,1) ; Po=Co*Pv`

ED1-7 Orifice, Sharp-Edged (Brooks 1993)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-7**



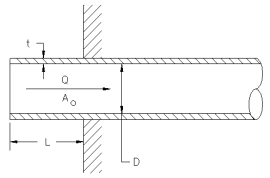
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Beta**	Diameter Ratio	(Dor/D)	(Dor/D)	2
Dor	Orifice Diameter	200	8	75–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**Co**	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Dor/D,1) ; Po=Co*Pv`

ED1-1 Duct Mounted in Wall (Idelchik 1986, Diagram 3-1)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-1**



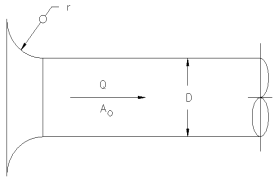
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
t	Wall Thickness	1.613	0.0635	0.319–3.51	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Extension	0	0	0–1500	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	700	1500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((L/D),(t/D),1) ; Po=Co*Pv`

ED1-2 Bellmouth, Standard Radius (Idelchik 1986, Diagram 3-4)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-2**



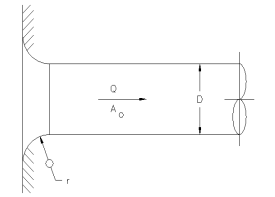
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	100–1200	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**r**	Bellmouth Radius	(r, mm)	(r, in.)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((r/D),2) ; Po=Co*Pv`

ED1-3 Bellmouth (Standard Radius) with Wall (Idelchik 1986, Diagram 3-4)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-3**



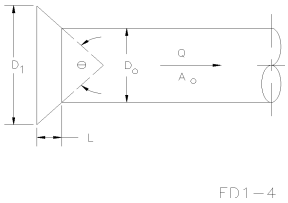
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	100–1200	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Q	Flow Rate	700	1500	5–9999999	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rho	Density	1.204	0.075	> 0	**r**	Bellmouth Radius	(r, mm)	(r, in.)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(r/D,2) ; Po=Co*Pv`

ED1-4 Conical Bellmouth/Sudden Contraction (Idelchik 1986, Diagram 3-6)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-4**



FD1—4

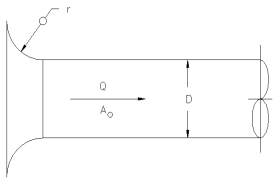
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	75–3000	**XL**	Length	(L, mm)	(L, in.)	0
Theta	Angle	60	60	0–180	**Theta 2**	Angle	(Theta, deg.)	(Theta, deg.)	0
L	Length	250	10	0–1500	**D1**	Diameter	(D1, mm)	(D1, in.)	0
Q	Flow Rate	700	1500	5–9999999	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rho	Density	1.204	0.075	> 0	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
					Co	Loss Coefficient	(Co)	(Co)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta2,(XL/Do),1) ; Po=Co*Pv`

ED1-8 Bellmouth, Short Radius (Idelchik 1986, Diagram 3-4)

ROUND • ENTRIES

Hierarchy Path: Exhaust/Return > Round > Entries > **ED1-8**



Var	Parameter	SI Def	IP Def	Range
D	Diameter	300	12	150–2150
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

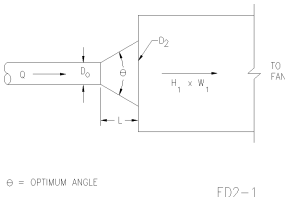
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
r	Bellmouth Radius	(r, mm)	(r, in.)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((r/D),1) ; Po=Co*Pv`

ED2-1 Conical Diffuser, Round to Plenum, Exhaust/Return Systems (Idelchik 1986, Diagram 5-8)

ROUND • PLENUMS

Hierarchy Path: Exhaust/Return > Round > Plenums > **ED2-1**



Var	Parameter	SI Def	IP Def	Range
Do	Diameter	300	12	75–3000
H1	Height	900	36	150–6000
W1	Width	900	36	150–6000
L	Length	600	24	0–1200
Q	Flow Rate	1000	2000	5–9999999
rho	Density	1.204	0.075	> 0

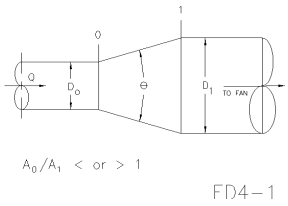
Var	Output Metric	SI Unit	IP Unit	Dec
Theta	Optimum Angle	(Theta, deg.)	(Theta, deg.)	0
D2	Diameter	(D2, mm)	(D2, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Do),(A1/Ao),1) ; Po=Co*Pvo`

ED4-1 Transition, Round to Round, Exhaust/Return Systems (Idelchik 1986, Diagrams 5-2 && 5-22)

ROUND • TRANSITIONS

Hierarchy Path: Exhaust/Return > Round > Transitions > **ED4-1**



Var	Parameter	SI Def	IP Def	Range
Do	Diameter	200	8	75–4000
D1	Diameter	300	12	75–4000
L	Length	300	12	0–1500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

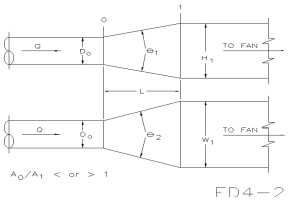
Var	Output Metric	SI Unit	IP Unit	Dec
Theta	Angle	(Theta, deg.)	(Theta, deg)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,(Ao/A1),1) ; Po=Co*Pvo`

ED4-2 Transition, Round to Rectangular, Exhaust/Return Systems (Idelchik 1986, Diagram 5-27)

ROUND • TRANSITIONS

Hierarchy Path: Exhaust/Return > Round > Transitions > **ED4-2**



Var	Parameter	SI Def	IP Def	Range
Do	Diameter	300	12	75–3000
H1	Height	450	18	75–4000
W1	Width	600	24	75–4000
L	Length	300	12	0–1500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

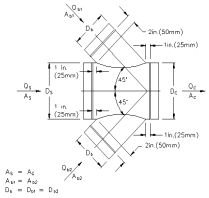
Var	Output Metric	SI Unit	IP Unit	Dec
Theta 1	Angle	(Theta1, deg.)	(Theta1, deg.)	0
Theta 2	Angle	(Theta2, deg.)	(Theta2, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,(Ao/A1),1) ; Po=Co*Pvo`

ED5-10 Double Wye, 45 Degree, Ab1 = Ab2, As = Ac, Converging (Idelchik 1986, Diagram 7-27)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > **ED5-10**



Var	Parameter	SI Def	IP Def	Range
Db	Diameter	150	6	25–2500
Dc	Diameter	300	12	50–2500
Qb1	Flow Rate	150	300	5–9999999
Qb2	Flow Rate	250	300	5–9999999
Qc	Flow Rate	750	1500	5–9999999
rho	Density	1.204	0.075	> 0

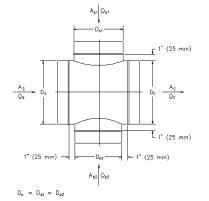
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn((Qs/Qc),(Qb2/Qb1),(Ab1/Ac),2) ; Pos=Cs*Pvs`

ED5-11 Cross, Ab1 = Ab2, As = Ac, Converging (Idelchik 1986, Diagram 7-29)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > **ED5-11**



Var	Parameter	SI Def	IP Def	Range
Db	Diameter	150	6	25–2500
Dc	Diameter	300	12	50–2500
Qb1	Flow Rate	150	300	5–9999999
Qb2	Flow Rate	250	300	5–9999999
Qc	Flow Rate	750	1500	5–9999999
rho	Density	1.204	0.075	> 0

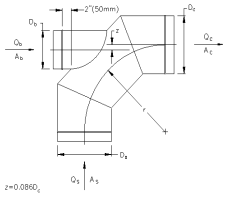
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn((Qs/Qc),2) ; Pos=Cs*Pvs`

ED5-8 Heel-Tapped Elbow, 90 Degree, r/Dc = 1.5, Converging (UMC 1985, Report SRF785E)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > **ED5-8**



Var	Parameter	SI Def	IP Def	Range
Ds	Diameter	275	10	50–2500
Db	Diameter	150	6	25–2500
Dc	Diameter	300	12	50–2500
Qs	Flow Rate	600	1200	5–9999999
Qb	Flow Rate	150	600	5–9999999
rho	Density	1.204	0.075	> 0

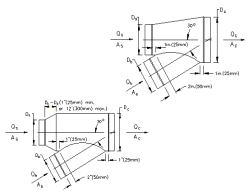
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),(As/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(Ab/Ac),(As/Ac),2)`

ED5-1 Wye, 30 Degree, Converging (Sepsy 1973)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > Wye > **ED5-1**



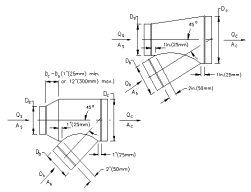
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ds	Diameter	275	11	50–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Dc	Diameter	300	12	50–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qs	Flow Rate	600	1200	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),(As/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(Ab/Ac),(As/Ac),2)`

ED5-2 Wye, 45 Degree, Converging (Sepsy 1973)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > Wye > **ED5-2**



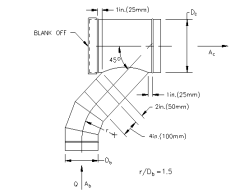
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Ds	Diameter	275	11	50–2500	**Vb**	{Branch }Velocity	(Vb, m/s)	(Vb, fpm)	1
Db	Diameter	150	6	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Dc	Diameter	300	12	50–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qs	Flow Rate	600	1200	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	5–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),(As/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(Ab/Ac),(As/Ac),2)`

ED5-6 Capped Wye, Branch with 45 Degree Elbow, Branch 90 Degrees (to Main, Converging (UMC 1983, Report SRN1083)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > Wye > **ED5-6**



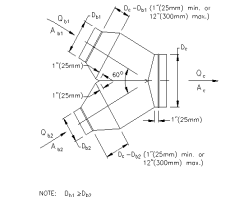
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Db	Diameter	150	6	25–2500	**Vb**	Velocity	(Vb, m/s)	Vb, fpm)	1
Dc	Diameter	300	12	50–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Q	Flow Rate	700	300	5–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvc**	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cb	Loss Coefficient	(Cb)	(Cb)	2
					Pob	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Ab/Ac),1) ; Pob=Cb*Pvb`

ED5-9 Symmetrical Wye, 60 Degree, Db1 > or = Db2, Converging (UMC 1985, Report SRF785E)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > Wye > **ED5-9**



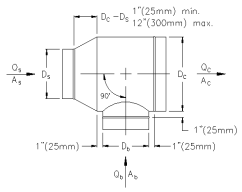
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Db1	Diameter	200	8	25–2500	**Vb1**	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Db2	Diameter	200	8	25–2500	**Pvb1**	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Dc	Diameter	300	12	50–2500	**Cb1**	Loss Coefficient	(Cb1)	(Cb1)	2
Qb1	Flow Rate	250	500	5–9999999	**Pob1**	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Qc	Flow Rate	750	1500	5–9999999	**Vb2**	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvb2**	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
					Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
					Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
					Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=(PI/4)*(Dc/ADC)^2 ; Ab1=(PI/4)*(Db1/ADC)^2 ; Ab2=(PI/4)*(Db2/ADC)^2`

ED5-3 Tee, Converging (Sepsy 1973)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > Tee > **ED5-3**



Var	Parameter	SI Def	IP Def	Range
Ds	Diameter	275	10	50–2500
Db	Diameter	150	6	25–2500
Dc	Diameter	300	12	50–2500
Qs	Flow Rate	600	1200	5–9999999
Qb	Flow Rate	125	250	5–9999999
rho	Density	1.204	0.075	> 0

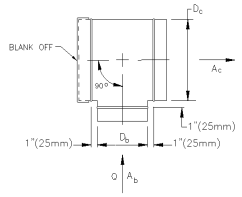
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=IFLE0((Dc-Dr),Cb1,Cb2) ; Cs=IFLE0((Dc-Dr),Cs1,Cs2) ; Pob=Cb*Pvb`

ED5-7 Capped Tee, Converging (UMC 1983, Report SRN1083)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > Tee > **ED5-7**



Var	Parameter	SI Def	IP Def	Range
Db	Diameter	150	6	25–2500
Dc	Diameter	300	12	50–2500
Q	Flow Rate	700	300	5–9999999
rho	Density	1.204	0.075	> 0

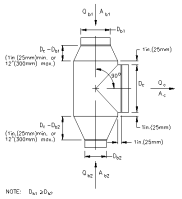
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Ab/Ac),1) ; Pob=Cb*Pvb`

ED5-4 Bullhead Tee without Vanes, Db1 > or = Db2, Converging (UMC 1985, Report SRF785E)

ROUND • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Round > Junctions, Converging > Tee > Bullhead > **ED5-4**



Var	Parameter	SI Def	IP Def	Range
Db1	Diameter	200	6	25–2500
Db2	Diameter	200	6	25–2500
Dc	Diameter	300	12	50–2500
Qb1	Flow Rate	250	500	5–9999999
Qc	Flow Rate	750	1200	5–9999999
rho	Density	1.204	0.075	> 0

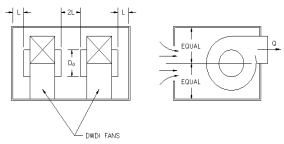
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=(PI/4)*(Dc/ADC)^2 ; Ab1=(PI/4)*(Db1/ADC)^2 ; Ab2=(PI/4)*(Db2/ADC)^2`

ED7-1 Centrifugal Fan Located in Plenum or Cabinet (AMCA 1990, Fig. 9-11)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > **ED7-1**



Var	Parameter	SI Def	IP Def	Range
Do	Diameter	300	12	100–2500
L	Length	100	4	0–6000
Q	Flow Rate	700	2500	5–9999999
rho	Density	1.204	0.075	> 0

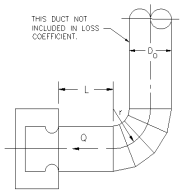
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(L/Do,1) ; Po=Co*Pv`

ED7-2 Fan Inlet, Centrifugal, SWSI, with 4-Gore Elbow (AMCA 1990, Fig. 9-5)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Centrifugal > **ED7-2**



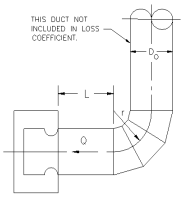
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	724	28.5	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rOD	Radius/Diameter Ra	1	1	0.5–3	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	600	24	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	6000	12000	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((L/Do),rOD,1) ; Po=Co*Pv`

ED7-3 Fan Inlet, Centrifugal, SWSI, with 3-Gore Elbow (AMCA 1990, Fig. 9-5)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Centrifugal > **ED7-3**



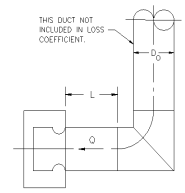
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	724	28.5	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
rOD	Radius/Diameter Ra	1	1	0.5–3	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
L	Length	600	24	0–6000	**Co**	Loss Coefficient	(Co)	(Co)	2
Q	Flow Rate	6000	12000	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

Governing Loss Formulation: `Co=fcn((L/Do),rOD,1) ; Po=Co*Pv`

ED7-4 Fan Inlet, Centrifugal, SWSI, with Mitered Elbow (AMCA 1990, Fig. 9-5)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Centrifugal > **ED7-4**



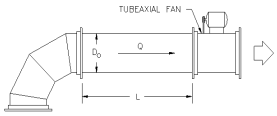
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	724	28.5	325–2600	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	600	24	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	6000	12000	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Po, Pa)	(Po, in. wg)	0

Governing Loss Formulation: `Co=fcn(L/Do,1) ; Po=Co*Pv`

ED7-5 Fan Inlet, Tubeaxial, 0.25 Hub-Tip Ratio, with 4-Gore Elbow (AMCA 1990, Fig. 9-2)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Tubeaxial > **ED7-5**



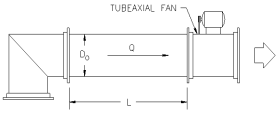
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	600	24	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Do),1) ; Po=Co*Pv`

ED7-6 Fan Inlet, Tubeaxial, 0.25 Hub-Tip Ratio, with Mitered Elbow (AMCA 1990, Fig. 9-2)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Tubeaxial > **ED7-6**



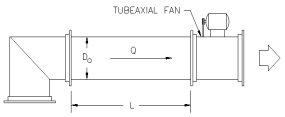
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	600	24	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(L1,1) ; Po=Co*Pv`

ED7-7 Fan Inlet, Tubeaxial, 0.35 Hub-Tip Ratio, with Mitered Elbow (AMCA 1990, Fig. 9-2)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Tubeaxial > **ED7-7**



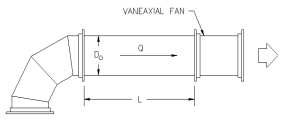
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	600	24	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Do),1) ; Po=Co*Pv`

ED7-8 Fan Inlet, Vaneaxial, 0.61 Hub-Tip Ratio, with 4-Gore Elbow (AMCA 1990, Fig. 9-2)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Vaneaxial > **ED7-8**



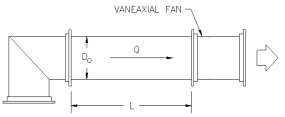
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	0	0	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Do),1) ; Po=Co*Pv`

ED7-9 Fan Inlet, Vaneaxial, 0.61 Hub-Tip Ratio, with Mitered Elbow (AMCA 1990, Fig. 9-2)

ROUND • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Round > Fan & System Interactions > Fan Inlet, Vaneaxial > **ED7-9**



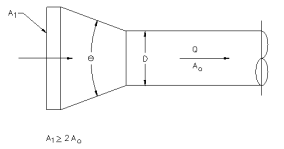
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
Do	Diameter	300	12	300–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	300	12	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	2500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Do),1) ; Po=Co*Pv`

ED10-1 Hood, Conical, Round Duct (Brandt 1946)

ROUND • HOODS

Hierarchy Path: Exhaust/Return > Round > Hoods > **ED10-1**



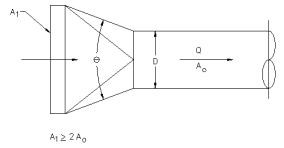
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Theta	Angle	90	90	0–180	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,1) ; Po=Co*Pv`

ED10-2 Hood, Rectangular-Shaped, Tapered, Round Duct (Brandt 1946)

ROUND • HOODS

Hierarchy Path: Exhaust/Return > Round > Hoods > **ED10-2**



Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
D	Diameter	300	12	75–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
Theta	Angle	90	90	0–180	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0

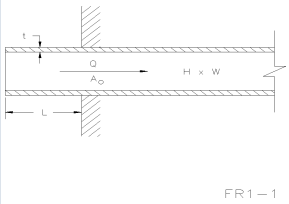
Governing Loss Formulation: `Co=fcn(Theta,1) ; Po=Co*Pv`

Exhaust/Return • Rectangular Duct Catalog (24 Fittings)

ER1-1 Duct Mounted in Wall (Idelchik 1986, Diagram 3-1)

RECTANGULAR • ENTRIES

Hierarchy Path: Exhaust/Return > Rectangular > Entries > **ER1-1**



FR1—1

Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	75–4000
W	Width	300	12	75–4000
t	Wall Thickness	1.613	0.0635	0.319–3.51
L	Extension	100	4	0–1500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

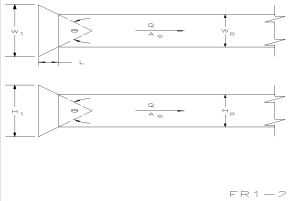
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Dh),(t/Dh),1) ; Po=Co*Pv`

ER1-2 Conical Bellmouth/Sudden Contraction (Idelchik 1986, Diagram 3-6)

RECTANGULAR • ENTRIES

Hierarchy Path: Exhaust/Return > Rectangular > Entries > **ER1-2**



FR1—2

Var	Parameter	SI Def	IP Def	Range
Ho	Height	300	12	75–4000
Wo	Width	450	18	75–4000
Theta	Angle	60	60	0–180
L	Length	250	10	0–1200
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

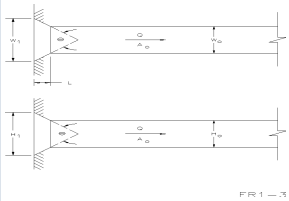
Var	Output Metric	SI Unit	IP Unit	Dec
XL	Length	(L, mm)	(L, in.)	0
Theta 3	Angle	(Theta, deg.)	(Theta, deg.)	0
H1	Height	(H1, mm)	(H1, in.)	0
W1	Width	(W1, mm)	(W1, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta3,XL/Dh,1) ; Po=Co*Pv`

ER1-3 Conical Bellmouth/Sudden Contraction, with Wall (Idelchik 1986, Diagram 3-7)

RECTANGULAR • ENTRIES

Hierarchy Path: Exhaust/Return > Rectangular > Entries > **ER1-3**



FR1—3

Var	Parameter	SI Def	IP Def	Range
Ho	Height	300	12	75–4000
Wo	Width	450	18	75–4000
Theta	Angle	60	60	0–180
L	Length	200	8	0–1200
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

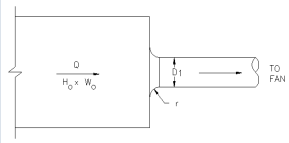
Var	Output Metric	SI Unit	IP Unit	Dec
XL	Length	(L, mm)	(L, in.)	0
Theta 2	Angle	(Theta, deg.)	(Theta, deg.)	0
H1	Height	(H1, mm)	(H1, in.)	0
W1	Width	(W1, mm)	(W1, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta2,XL/Dh,1) ; Po=Co*Pv`

ER2-1 Bellmouth, Plenum to Round, Exhaust/Return Systems (Idelchik 1986, Diagram 4-9)

RECTANGULAR • PLENUMS

Hierarchy Path: Exhaust/Return > Rectangular > Plenums > **ER2-1**



Var	Parameter	SI Def	IP Def	Range
D1	Diameter	300	12	75–3000
Ho	Plenum Height	900	36	150–6000
Wo	Plenum Width	900	36	150–6000
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

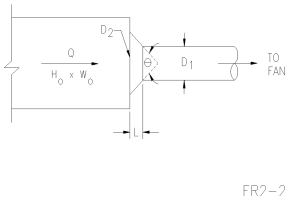
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
y	Bellmouth Radius	(r, mm)	(r, in.)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=C1*Pv1/Pvo ; Po=Co*Pvo`

ER2-2 Conical Bellmouth/Sudden Contraction, Plenum to Round (Exhaust/Return Systems (Idelchik 1986, Diagram 4-9))

RECTANGULAR • PLENUMS

Hierarchy Path: Exhaust/Return > Rectangular > Plenums > **ER2-2**



FR2-2

Var	Parameter	SI Def	IP Def	Range
D1	Diameter	600	24	75–3000
Ho	Plenum Height	6250	250	150–12500
Wo	Plenum Width	6250	250	150–12500
Theta	Angle	60	60	0–180
L	Length	175	7	0–1500
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

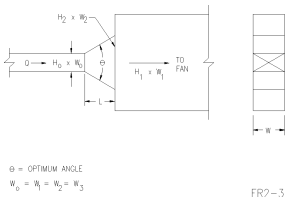
Var	Output Metric	SI Unit	IP Unit	Dec
XL	Length	(L, mm)	(L, in.)	0
Theta 2	Angle	(Theta, deg.)	(Theta, deg.)	0
D2	Diameter	(D2, mm)	(D2, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=C1*Pv1/Pvo ; Po=Co*Pvo`

ER2-3 Plain Diffuser (Two Sides Parallel), Rectangular to Plenum (Exhaust/Return Systems (Idelchik 1986, Diagram 5-10))

RECTANGULAR • PLENUMS

Hierarchy Path: Exhaust/Return > Rectangular > Plenums > **ER2-3**



FR2-3

Var	Parameter	SI Def	IP Def	Range
Ho	Height	300	12	75–4000
Wo	Width	300	12	75–4000
H1	Plenum Height	1200	48	150–12500
L	Length	300	12	0–1200
Q	Flow Rate	1250	2500	5–9999999
rho	Density	1.204	0.075	> 0

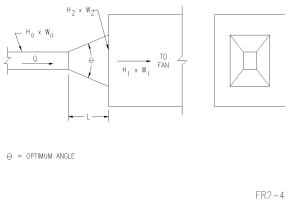
Var	Output Metric	SI Unit	IP Unit	Dec
Theta	Optimum Angle	(Theta, deg.)	(Theta, deg.)	0
H2	Height	(H2, mm)	(H2, in.)	0
W2	Width	(W2, mm)	(W2, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Ho),(A1/Ao),1) ; Po=Co*Pvo`

ER2-4 Pyramidal Diffuser, Rectangular to Plenum (Exhaust/Return Systems (Idelchik 1986, Diagram 5-9))

RECTANGULAR • PLENUMS

Hierarchy Path: Exhaust/Return > Rectangular > Plenums > **ER2-4**



FR2-4

Var	Parameter	SI Def	IP Def	Range
Ho	Height	300	12	75–4000
Wo	Width	300	12	75–4000
H1	Height	600	24	1–12500
W1	Width	600	24	1–12500
L	Length	300	12	0–1200
Q	Flow Rate	1000	2000	5–9999999
rho	Density	1.204	0.075	> 0

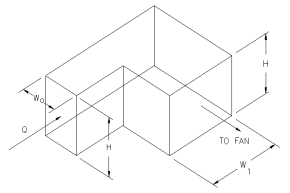
Var	Output Metric	SI Unit	IP Unit	Dec
Theta	Optimum Angle	(Theta, deg.)	(Theta, deg.)	0
H2	Plenum Height	(H2, mm)	(H2, in.)	0
W2	Plenum Width	(W2, mm)	(W2, in.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((L/Dh),(A1/Ao),1) ; Po=Co*Pvo`

ER3-1 Elbow, 90 Degree, Variable Inlet/Outlet Areas (Exhaust/ Return Systems (Idelchik 1986, Diagram 6-4))

RECTANGULAR • ELBOWS

Hierarchy Path: Exhaust/Return > Rectangular > Elbows > **ER3-1**



Var	Parameter	SI Def	IP Def	Range
H	Height	250	10	75–4000
Wo	Width	250	10	75–4000
W1	Width	500	20	75–4000
Q	Flow Rate	700	1500	5–9999999
rho	Density	1.204	0.075	> 0

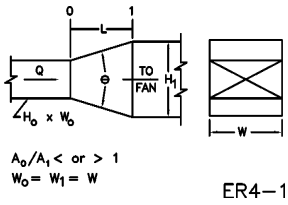
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn((W1/Wo),(H/Wo),1) ; Po=Co*Pvo`

ER4-1 Transition, Rectangular, Two Sides Parallel, Symmetrical (Exhaust/Return Systems (Idelchik 1986, Diagram 5-5))

RECTANGULAR • TRANSITIONS

Hierarchy Path: Exhaust/Return > Rectangular > Transitions > **ER4-1**



ER4−1

Var	Parameter	SI Def	IP Def	Range
Ho	Height	300	12	75–4000
Wo	Width	400	16	75–4000
H1	Height	450	18	75–4000
L	Length	600	24	0–1500
Q	Flow Rate	1000	2000	5–9999999
rho	Density	1.204	0.075	> 0

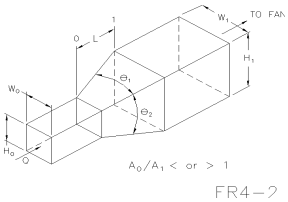
Var	Output Metric	SI Unit	IP Unit	Dec
Theta	Angle	(Theta, deg.)	(Theta, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,(Ao/A1),1) ; Po=Co*Pvo`

ER4-2 Transition, Pyramidal, Exhaust/Return Systems (Idelchik 1986, Diagram 5-4)

RECTANGULAR • TRANSITIONS

Hierarchy Path: Exhaust/Return > Rectangular > Transitions > **ER4-2**



FR4−2

Var	Parameter	SI Def	IP Def	Range
Ho	Height	300	12	75–4000
Wo	Width	300	12	75–4000
H1	Height	450	18	75–4000
W1	Width	450	18	75–4000
L	Length	600	24	0–1500
Q	Flow Rate	1000	2000	5–9999999
rho	Density	1.204	0.075	> 0

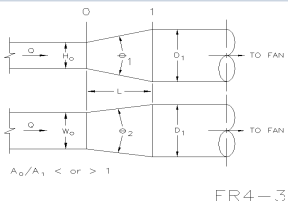
Var	Output Metric	SI Unit	IP Unit	Dec
Theta 1	Angle	(Theta1, deg.)	(Theta1, deg.)	0
Theta 2	Angle	(Theta2, deg.)	(Theta2, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,(Ao/A1),1) ; Po=Co*Pvo`

ER4-3 Transition, Rectangular to Round, Exhaust/Return Systems (Idelchik 1986, Diagram 5-27)

RECTANGULAR • TRANSITIONS

Hierarchy Path: Exhaust/Return > Rectangular > Transitions > **ER4-3**



FR4−3

Var	Parameter	SI Def	IP Def	Range
Ho	Height	300	12	75–4000
Wo	Width	300	12	75–4000
D1	Diameter	600	24	75–3000
L	Length	600	24	0–1500
Q	Flow Rate	750	1500	5–9999999
rho	Density	1.204	0.075	> 0

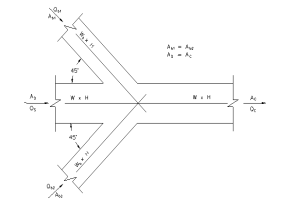
Var	Output Metric	SI Unit	IP Unit	Dec
Theta 1	Angle	(Theta1, deg.)	(Theta1, deg.)	0
Theta 2	Angle	(Theta2, deg.)	(Theta2, deg.)	0
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
V1	Velocity	(V1, m/s)	(V1, fpm)	1
Pvo	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
Pv1	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
C1	Loss Coefficient	(C1)	(C1)	2
Po	Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn(Theta,(Ao/A1),1) ; Po=Co*Pvo`

ER5-9 Double Wye of the Type As = Ac, Ab1 = Ab2, 45 Degree, Converging (Idelchik 1986, Diagram 7-27)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > **ER5-9**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Wb	Width	150	6	25–2500
Qb1	Flow Rate	150	300	5–9999999
Qb2	Flow Rate	150	300	5–9999999
Qc	Flow Rate	750	1500	5–9999999
rho	Density	1.204	0.075	> 0

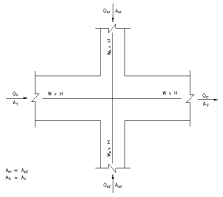
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	Branch 1 Pressure Loss	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	Branch 2 Pressure Loss	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	Main Pressure Loss	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn[(Qs/Qc),(Qb2/Qb1),(Ab1/Ac),2] ; Pos=Cs*Pvs`

ER5-10 Cross of the Type As = Ac, Ab1 = Ab2, Converging (Idelchik 1986, Diagram 7-29)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > **ER5-10**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Wb	Width	150	6	25–2500
Qb1	Flow Rate	250	300	5–9999999
Qb2	Flow Rate	150	300	5–9999999
Qc	Flow Rate	750	1500	5–9999999
rho	Density	1.204	0.075	> 0

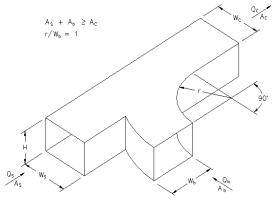
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	**Pressure Loss**	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cs=fcn[(Qs/Qc),2] ; Pos=Cs*Pvs`

ER5-1 Smooth Radius Wye of the Type As + Ab > or = Ac, Branch 90 (Degrees to Main, Converging (Idelchik 1986, Diagram 7-11)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Wye > **ER5-1**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
Ws	Width	250	10	25–2500
Wb	Width	150	6	25–2500
Wc	Width	300	12	25–2500
Qs	Flow Rate	600	1200	5–9999999
Qb	Flow Rate	150	300	5–9999999
rho	Density	1.204	0.075	> 0

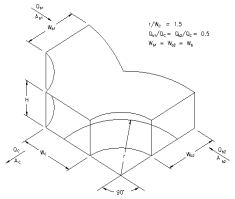
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn((Qb/Qc),(Ab/Ac),(As/Ac),1) ; Pob=Cb*Pvb ; Cs=fcn((Qs/Qc),(Ab/Ac),(As/Ac),2)`

ER5-4 Wye, Symmetrical, Dovetail, Qb/Qc = 0.5, Converging (Idelchik 1986, Diagram 7-24)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Wye > **ER5-4**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
Wb	Width	150	6	25–2500
Wc	Width	300	12	25–2500
Qc	Flow Rate	750	1500	5–9999999
rho	Density	1.204	0.075	> 0

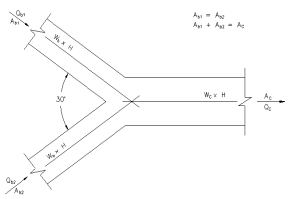
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vc, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Cb=fcn((Ab/Ac),1) ; Pob=Cb*Pvb`

ER5-6 Symmetrical Wye Ab1 + Ab2 = Ac, Ab1 = Ab2, 30 Degree, Converging (Idelchik 1986, Diagram 7-30)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Wye > **ER5-6**



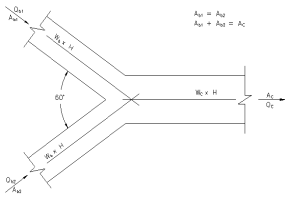
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb1**	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Wb	Width	150	6	25–2500	**Pvb1**	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Wc	Width	300	12	25–2500	**Cb1**	Loss Coefficient	(Cb1)	(Cb1)	2
Qb1	Flow Rate	400	800	0–9999999	**Pob1**	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Qc	Flow Rate	750	1500	5–9999999	**Vb2**	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvb2**	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
					Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
					Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
					Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=H*Wc/ARC ; Ab=H*Wb/ARC ; Qb2=Qc-Qb1`

ER5-7 Symmetrical Wye Ab1 + Ab2 = Ac, Ab1 = Ab2, 60 Degree, Converging (Idelchik 1986, Diagram 7-30)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Wye > **ER5-7**



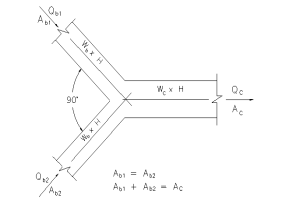
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb1**	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Wb	Width	150	6	25–2500	**Pvb1**	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Wc	Width	300	12	25–2500	**Cb1**	Loss Coefficient	(Cb1)	(Cb1)	2
Qb1	Flow Rate	400	800	0–9999999	**Pob1**	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Qc	Flow Rate	750	1500	5–9999999	**Vb2**	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvb2**	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
					Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
					Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
					Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=H*Wc/ARC ; Ab=H*Wb/ARC ; Qb2=Qc-Qb1`

ER5-8 Symmetrical Wye of the Type Ab1 + Ab2 = Ac, Ab1 = Ab2 (90 Degree, Converging (Idelchik 1986, Diagrams 7-30)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Wye > **ER5-8**



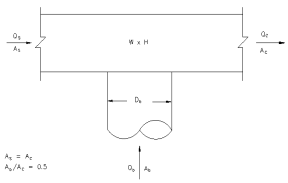
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb1**	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Wb	Width	150	6	25–2500	**Pvb1**	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Wc	Width	300	12	25–2500	**Cb1**	Loss Coefficient	(Cb1)	(Cb1)	2
Qb1	Flow Rate	400	800	0–9999999	**Pob1**	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Qc	Flow Rate	750	1500	5–9999999	**Vb2**	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvb2**	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
					Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
					Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
					Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=H*Wc/ARC ; Ab=H*Wb/ARC ; Qb2=Qc-Qb1`

ER5-2 Tee, Round Tap to Rectangular Main [Ab/As=Ab/Ac=0.5; As/Ac], Converging (SMACNA 1990, Sepsy 1973)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Tee > **ER5-2**



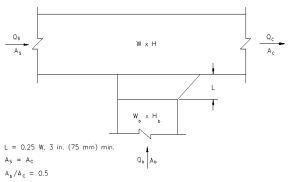
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height	300	12	25–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
W	Width	300	12	25–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Db	Diameter	150	6	25–2500	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Qs	Flow Rate	600	1200	5–9999999	**Pob**	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Qb	Flow Rate	150	300	0–9999999	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
rho	Density	1.204	0.075	> 0	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
					Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),2]`

ER5-3 Tee, 45 Degree Entry Branch [Ab/As=Ab/Ac=0.5; As=Ac], Converging (SMACNA 1990, Sepsy 1973)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Tee > **ER5-3**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
W	Width	300	12	25–2500
Hb	Height	150	6	25–2500
Wb	Width	150	6	25–2500
Qs	Flow Rate	600	1200	5–9999999
Qb	Flow Rate	150	300	0–9999999
rho	Density	1.204	0.075	> 0

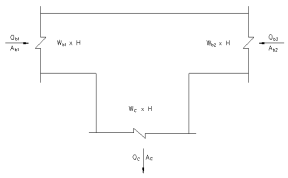
Var	Output Metric	SI Unit	IP Unit	Dec
Vb	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
Pvb	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in. wg)	0
Cb	Loss Coefficient	(Cb)	(Cb)	2
Pob	**Branch Pressure Loss**	(Pa)	(in. wg)	0
Vs	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Vc	Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvs	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
Cs	Loss Coefficient	(Cs)	(Cs)	2
Pos	**Main Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Cb=fcn[(Qb/Qc),1] ; Pob=Cb*Pvb ; Cs=fcn[(Qs/Qc),2]`

ER5-5 Bullhead Tee without Vanes, Converging (Idelchik 1986, Diagram 7-23)

RECTANGULAR • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Rectangular > Junctions, Converging > Tee > **ER5-5**



Var	Parameter	SI Def	IP Def	Range
H	Height	300	12	25–2500
Wb1	Width	200	8	25–2500
Wb2	Width	200	8	25–2500
Wc	Width	300	12	25–2500
Qb1	Flow Rate	400	800	5–9999999
Qc	Flow Rate	750	1500	5–9999999
rho	Density	1.204	0.075	> 0

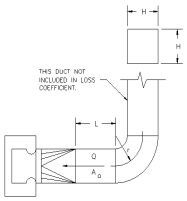
Var	Output Metric	SI Unit	IP Unit	Dec
Vb1	{Branch 1}Velocity	(Vb1, m/s)	(Vb1, fpm)	1
Pvb1	Vel Pres at Vb1	(Pvb1, Pa)	(Pvb1, in. wg)	0
Cb1	Loss Coefficient	(Cb1)	(Cb1)	2
Pob1	**Branch 1 Pressure Loss**	(Pa)	(in. wg)	0
Vb2	{Branch 2}Velocity	(Vb2, m/s)	(Vb2, fpm)	1
Pvb2	Vel Pres at Vb2	(Pvb2, Pa)	(Pvb2, in. wg)	0
Cb2	Loss Coefficient	(Cb2)	(Cb2)	2
Pob2	**Branch 2 Pressure Loss**	(Pa)	(in. wg)	0
Vc	{Main}Velocity	(Vc, m/s)	(Vc, fpm)	1
Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0

Governing Loss Formulation: `Ac=H*Wc/ARC ; Ab1=H*Wb1/ARC ; Ab2=H*Wb2/ARC`

ER7-1 Fan Inlet, Centrifugal, SWSI, 90 Degree Smooth Radius Elbow (Square) (AMCA 1990, Fig. 9-6)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Rectangular > Fan & System Interactions > Fan Inlet, Centrifugal, SWSI > **ER7-1**



Var	Parameter	SI Def	IP Def	Range
H	Height and Width	375	15	325–3000
L	Length	300	12	0–6000
rOH	Radius/Height Rati	1	1	0.5–2
Q	Flow Rate	700	2500	5–9999999
rho	Density	1.204	0.075	> 0

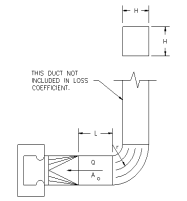
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[(L/H),rOH,1] ; Po=Co*Pv`

ER7-2 Fan Inlet, Centrifugal, SWSI, 90 Deg. Smooth Radius Elbow (Square) with Fulll Radius Vanes (ACMA 1990, Fig. 9-6)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Rectangular > Fan & System Interactions > Fan Inlet, Centrifugal, SWSI > **ER7-2**



Var	Parameter	SI Def	IP Def	Range
H	Height and Width	375	15	325–3000
L	Length	300	12	0–6000
rOH	Radius/Height Rati	1	1	0.5–2
Q	Flow Rate	700	2500	5–9999999
rho	Density	1.204	0.075	> 0

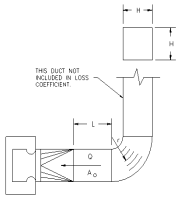
Var	Output Metric	SI Unit	IP Unit	Dec
Vo	Velocity	(Vo, m/s)	(Vo, fpm)	1
Pv	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
Co	Loss Coefficient	(Co)	(Co)	2
Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=fcn[(L/H),rOH,1] ; Po=Co*Pv`

ER7-3 Fan Inlet, Centrifugal, SWSI, 90 Degree Smooth Radius Elbow (Square) with Short Vanes (AMCA 1990, Fig. 9-6)

RECTANGULAR • FAN & SYSTEM INTERACTIONS

Hierarchy Path: Exhaust/Return > Rectangular > Fan & System Interactions > Fan Inlet, Centrifugal, SWSI > **ER7-3**



Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
H	Height and Width	375	15	325–3000	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
L	Length	300	12	0–6000	**Pv**	Vel Pres at Vo	(Pv, Pa)	(Pv, in. wg)	0
rOH	Radius/Height Rati	1	1	0.5–2	**Co**	Los Ceofficient	(Co)	(Co)	2
Q	Flow Rate	700	2500	5–9999999	**Po**	**Pressure Loss**	(Pa)	(in. wg)	0
rho	Density	1.204	0.075	> 0					

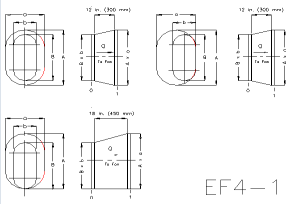
Governing Loss Formulation: `Co=fcn[(L/H),rOH,1] ; Po=Co*Pv`

Exhaust/Return • Flat Oval Duct Catalog (3 Fittings)

EF4-1 Transition, Flat Oval, Concentric && Eccentrix, Exhaust/Return Air Systems (Zhang, Sun, Ford && Denny 2014)

FLAT OVAL • TRANSITIONS

Hierarchy Path: Exhaust/Return > Flat Oval > Transitions > **EF4-1**



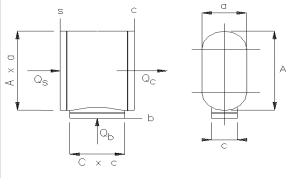
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
B	Upstream Major Axi	300	12	70–2500	**Vo**	Velocity	(Vo, m/s)	(Vo, fpm)	1
b	Upstream Miinor Ax	200	8	70–2500	**V1**	Velocity	(V1, m/s)	(V1, fpm)	1
A	Downstream Major A	600	24	70–3500	**Pvo**	Vel Pres at Vo	(Pvo, Pa)	(Pvo, in. wg)	0
a	Downstream Minor A	400	16	70–2500	**Pv1**	Vel Pres at V1	(Pv1, Pa)	(Pv1, in. wg)	0
Q	Flow Rate	700	1500	5–9999999	**Co**	Loss Coefficient	(Co)	(Co)	2
rho	Density	1.204	0.075	> 0	**C1**	Loss Coefficient	(C1)	(C1)	2
					Po	**Pressure Loss**	(Pa)	(in. wg)	0

Governing Loss Formulation: `Co=C1*Pv1/Pvo ; Po=Pvo*Co`

EF5-1 Tee, Flat Oval, Straight Body Branch, Converging (Kulkarni, Cui && Idem 2011)

FLAT OVAL • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Flat Oval > Junctions, Converging > **EF5-1**



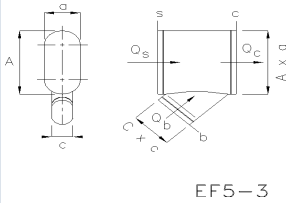
Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Main, Major Axis	455	18	70–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
H	Main, Minor Axis	255	10	70–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in.wg)	0
Qc	Flow Rate	1250	2600	5–9999999	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Wb	Branch, Major Axis	230	10	70–2500	**Pob**	**Branch Pressure Loss**	(Pob. Pa)	(Pob, in. wg)	0
Hb	Branch, Minor Axis	100	5	70–2500	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Qb	Flow Rate	150	260	0–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pos, Pa)	(Pos, in. wg)	0

Governing Loss Formulation: `Cb=IFLE0(Nb, -100, 0.623*Xb+1.559) ; Pob=Pvb*Cb ; TCs=IFLE0(Ns,100,-2.561*Xs-1.683`

EF5-3 Wye, Flat Oval, 45 Degree, Straight Body Branch, Converging (Kulkarni, Cui && Idem 2011)

FLAT OVAL • JUNCTIONS, CONVERGING

Hierarchy Path: Exhaust/Return > Flat Oval > Junctions, Converging > **EF5-3**



Var	Parameter	SI Def	IP Def	Range	Var	Output Metric	SI Unit	IP Unit	Dec
W	Main, Major Axis	455	18	70–2500	**Vb**	{Branch}Velocity	(Vb, m/s)	(Vb, fpm)	1
H	Main, Minor Axis	255	10	70–2500	**Pvb**	Vel Pres at Vb	(Pvb, Pa)	(Pvb, in.wg)	0
Qc	Flow Rate	1250	2600	5–9999999	**Cb**	Loss Coefficient	(Cb)	(Cb)	2
Wb	Branch, Major Axis	230	10	70–2500	**Pob**	**Branch Pressure Loss**	(Pob, Pa)	(Pob, in. wg)	0
Hb	Branch, Minor Axis	100	5	70–2500	**Vs**	{Main}Velocity	(Vs, m/s)	(Vs, fpm)	1
Qb	Flow Rate	150	260	0–9999999	**Vc**	Velocity	(Vc, m/s)	(Vc, fpm)	1
rho	Density	1.204	0.075	> 0	**Pvs**	Vel Pres at Vs	(Pvs, Pa)	(Pvs, in. wg)	0
					Pvc	Vel Pres at Vc	(Pvc, Pa)	(Pvc, in. wg)	0
					Cs	Loss Coefficient	(Cs)	(Cs)	2
					Pos	**Main Pressure Loss**	(Pos, Pa)	(Pos, in. wg)	0

Governing Loss Formulation: `Cb=IFLE0(Nb,-100,0.563*X1+1.304) ; Pob=Cb*Pvb ; TCs=0.692*Xs+1.075`